Project Report On

Design And Implementation Of Zero Trust Micro-VPN For Campus Networks

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Computer Science and Engineering

By

Abhirami C U (U2203010)

Alan Suresh Joseph (U2203023)

Aparna Velickakathu Sunil (U2203051)

Aryasree Nambiar (U2203054)

Under the guidance of

Mr. Sandy Joseph

Department of Computer Science and Engineering
Rajagiri School of Engineering & Technology (Autonomous)
(Parent University: APJ Abdul Kalam Technological University)
Rajagiri Valley, Kakkanad, Kochi, 682039

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CERTIFICATE

This is to certify that the project report entitled ***"Design And Implementation Of Zero Trust Micro -VPN For Campus Networks"*** *is a bonafide record of the work done by* ***Abhirami C U (U2203010), Alan Suresh Joseph (U2203023), Aparna Velickakathu Sunil (U2203051), Aryasree Nambiar (U2203054)****, submitted to the Rajagiri School of Engineering & Technology (RSET) (Autonomous) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B. Tech.) in "Computer Science and Engineering" during the academic year 2025-2026.*

Project Guide

Mr. Sandy Joseph

Assistant Professor

Dept. of CSE

RSET

Project Co-ordinator

Dr. Jisha G

Associate Professor

Dept. of CSE

RSET

Head of the Department

Dr. Preetha K G

Professor

Dept. of CSE

RSET

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Abhirami C U

Alan Suresh Joseph

Aparna Velickakathu Sunil

Aryasree Nambiar

Abstract

The growing use of remote access, cloud-based services, and distributed user environments in modern campus networks has made it crucial to ensure secure communication and granular access control. In order to protect against contemporary threats like credential misuse, unauthorised access, session hijacking, and lateral movement within institutional systems, traditional perimeter-based security models that rely on implicit trust and static firewall protections are insufficient. The design and implementation of an Academic Access Management System with Zero Trust capabilities combined with a safe Micro-VPN architecture for campus settings are presented in this study. The proposed system enforces continuous verification, identity-aware access control, and encrypted communication to secure interactions with a centralized academic management portal used by students, parents, faculty, and administrators. Using AES for effective data encryption and RSA for safe key exchange, the architecture combines a unique secure VPN tunnel with hybrid cryptographic protection. Strict role-based access control (RBAC), JSON Web Token (JWT) session management, and Google Authenticator-based multi-factor authentication (TOTP) all improve authentication and authorisation. Before allowing access to protected academic resources, a dynamic Zero Trust policy engine continuously verifies user identity, role privileges, and contextual parameters. Along with centralised academic management features like class configuration, student enrolment, faculty assignment, academic record management, and grievance handling, the system also supports microsegmented portals for students, parents, faculty, and administrators. To guarantee accountability and traceability, secure audit logging is used to document all authentication events, policy choices, and system operations. In order to replicate actual campus settings, the system is deployed on a cloud platform using a web-based architecture with a PostgreSQL database backend.

Keywords: Zero Trust Architecture, Micro-VPN, Multi-Factor Authentication, JWT Authentication, Role-Based Access Control, Hybrid Encryption (AES-RSA), Academic Management Systems, Network Security.

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List of Abbreviations

VPN - Virtual Private Network

OTP - One Time Password

AES - Advanced Encryption Standard

RSA - Rivest–Shamir–Adleman

JWT - JSON Web Token

ZTA - Zero Trust Architecture

PDP - Policy Decision Point

PEP - Policy Enforcement Point

ZTNA - Zero Trust Network Access

IOT - Internet of Things

MFA - Multi Factor Authentication

RBAC - Role-Based Access Control

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Chapter 1

Introduction

The Zero Trust security concept and its implementation in campus networks are introduced in this chapter. It gives a brief synopsis of the chapter after outlining the report's history, problem statement, scope, objectives, obstacles, assumptions, societal relevance, and organization. The main objective is to offer a solid basis for comprehending the necessity of secure, role-based, and continually validated network access in contemporary college settings.

1.1 Background

As educational institutions increasingly adopt digital devices and online platforms, campus networks are becoming more intricate and vulnerable to unauthorized access and insider attacks. Conventional perimeter-focused security frameworks, which depend on firewalls and VPNs, operate on the premise that users inside the network are trustworthy and concentrate mainly on external threats. Yet, the proliferation of endpoints, cloud services, and mobile devices has made this strategy inadequate.

The Zero Trust security paradigm overturns this assumption by requiring ongoing verification of user identities, enforcing stringent access controls, and maintaining minimal—essentially "zero"—implicit trust. Deploying a micro VPN within a campus network supports secure, encrypted connections between users and network assets while following Zero Trust guidelines. In doing so, it helps ensure that sensitive information, including student records, faculty data, and administrative documents, is safeguarded against unauthorized access. [**?**].

1.2 Problem Definition

The main objective of this project is to develop and deploy a secure Zero Trust Micro VPN tailored for campus networks, which continuously authenticates user identities, applies role-based access controls, and safeguards critical resources. It targets the weaknesses of current campus network infrastructures, particularly their exposure to insider threats, unauthorized access, and unencrypted data transmissions. The proposed system integrates secure VPN communication, role-based portals for academic management, and modern authentication mechanisms to ensure that only verified users can access sensitive institutional resources.

1.3 Scope and Motivation

Scope: The scope of this project involves creating a secure VPN-enabled academic management system for campus environments that adheres to Zero Trust principles. It includes dedicated portals for students and parents, separate interfaces for faculty and administrators, multi-factor authentication using Google Authenticator (TOTP), detailed access logging, and role-specific permission controls. The system also incorporates JWT-based session management, hybrid encryption using AES and RSA, and a PostgreSQL database for structured storage of academic and security-related data. The system is designed to safeguard sensitive academic and administrative information while maintaining an intuitive user experience. It also establishes a foundation for adding additional network security components in subsequent development phases.

Motivation: The driving force behind this project is the need to strengthen campus network security beyond traditional perimeter-based protections. As cyber threats and insider attacks increase, it becomes critical to continuously validate every user request and apply access controls in a dynamic way. Adopting a Zero Trust model with a micro VPN helps maintain data confidentiality, integrity, and availability, while also showcasing the real-world implementation of advanced network security principles in academic environments.

1.4 Objectives

- Design a micro VPN solution that applies Zero Trust principles within campus networks, demonstrated through a role-based academic management system.
- Implement robust authentication methods, including multi-factor authentication using Google Authenticator and secure session management.
- Create role-based access control policies for students, parents, faculty, and administrative staff.
- Preserve comprehensive access logs to track user actions, identify suspicious behavior, and uphold security policies.
- Safeguard confidential information such as student grades, attendance records, and administrative data using encrypted communication.
- Deliver a scalable and extensible architecture that supports cloud deployment, secure database management, and simultaneous access by multiple users.

1.5 Challenges

Ensuring real-time user identity verification and implementing secure multi-factor authentication mechanisms are two of the project's main challenges. Managing the complexity of role-based access control in a dynamic campus context and preventing unauthorized access while preserving usability are further difficulties. Additionally, integrating secure VPN communication, hybrid encryption mechanisms, and database management while maintaining system performance presents technical challenges.

1.6 Assumptions

The project is predicated on: Every user has a registered account with valid authentication credentials.

- Secure routing and VPN installations are supported by the campus network architecture or cloud deployment environment.

- Users will adhere to authentication requirements and security standards when accessing the academic management system.

1.7 Societal / Industrial Relevance

This endeavor is relevant to both industry and society. It improves the security of private student, faculty, and administrative data for educational institutions through the implementation of Zero Trust principles, encrypted communication, and strict access control mechanisms. The architecture, which can be tailored to corporate networks, governmental organizations, and enterprise settings needing secure communication and identity-aware access control, shows the industry how Zero Trust concepts and micro VPN deployment are practically implemented.

1.8 Organization of the Report

The format of the report is as follows:

- **Chapter 1: Introduction** –This chapter covers the following topics: background, problem definition, scope and motivation, objectives, risks and challenges, assumptions, societal/industrial significance, report organization, and chapter summary.
- **Chapter 2: Literature Survey** –This chapter includes an introduction, a review of relevant papers, a comparison of approaches, a list of gaps, and a summary of the chapter.
- **Chapter 3: System Design** – This chapter covers the general system architecture, design diagrams, component design, algorithms used, software and hardware requirements, datasets identified, database, budget (if any), module division and work responsibilities, important deliverables, project timeline, UI design (if any), and chapter summary.
- **Chapter 4: Conclusion** – This chapter contains a recap of the chapter as well as a plan for the upcoming semester.

1.9 Summary of Chapter

By describing the need for improved security in campus networks, identifying the issue, laying out the goals, and emphasizing the applicability of the suggested Zero Trust Micro VPN solution, this chapter laid the groundwork for the project. The literature, system design, implementation, and assessment will all be covered in detail in the upcoming chapters.

Chapter 2

Literature Survey

Zero Trust Architecture (ZTA) has been developed as a new paradigm of cybersecurity that removes implicit trust from the network infrastructure of enterprises. In traditional security architecture, implicit trust is given to the nodes within the network infrastructure of enterprises, which has made security systems vulnerable to attacks like insider attacks and lateral movement attacks. Zero Trust Architecture has overcome the challenges associated with traditional security architecture. This chapter discusses the important research work associated with Zero Trust Architecture and trust evaluation mechanisms.

2.1 Rose, S., Borchert, O., Mitchell, S., & Connelly, S. (2020). Zero Trust Architecture. NIST Special Publication 800-207 [1]

In the research article by Rose et al. (2020), the authors presented the Zero Trust Architecture framework that explains the fundamental concepts of Zero Trust security. The architecture also includes key components like Policy Decision Point (PDP) and Policy Enforcement Point (PEP) for implementing dynamic policy control. The architecture also changes the traditional security paradigm to identity-based or device-based access control.

This research article offers a good foundation for implementing a Zero Trust system. However, the article lacks implementation guidelines. Moreover, the need for continuous verification might add computational overhead to the system.

2.2 Ward, R., & Beyer, B. (2014). BeyondCorp: A New Approach to Enterprise Security [2]

Ward and Beyer proposed the Google BeyondCorp architecture, which presents a practical example of the application of Zero Trust principles. This architecture eliminates the idea

of a trusted network, where each request is verified based on the identity of the user. The BeyondCorp architecture is a successful example of the implementation of the Zero Trust model for enterprise infrastructure. However, the implementation of this model is associated with high infrastructure automation and identity management, which may add complexity to the infrastructure.

2.3 Kumar, A., et al. (2021). A Comprehensive Review of Zero Trust Network Architecture (ZTNA) and Deployment Frameworks [3]

A comprehensive survey on the different Zero Trust Network Architecture models, along with deployment frameworks, has been presented by Kumar et al. In the research, the different architectural strategies, such as broker-based, agent-based, and software-defined perimeter-based, are discussed.

The research has been quite informative regarding the implementation strategies, along with the major challenges that need to be resolved during deployment. However, the research is more theoretical, with no experimental results being discussed.

2.4 Jøsang, A., Ismail, R., & Boyd, C. (2007). A Survey of Trust and Reputation Systems for Online Service Provision [4]

In the research conducted by Jøsang et al., the authors introduced the trust and reputation models for online service environments. In the research, the authors proposed mathematical methods for computing trust scores based on the aggregation of weighted positive and negative events during interaction.

Although the research has a high theoretical potential for trust computation, it was originally intended for use in the reputation system, not the access control environment.

2.5 Gambo, M. L., & Almulhem, A. (2025). A Systematic Literature Review on the Implementation and Challenges of Zero Trust Architecture Across Domains [5]

In the systematic literature review conducted by Gambo and Almulhem, the implementation of Zero Trust Architecture was analyzed for different domains such as healthcare,

cloud computing, Internet of Things, etc. The authentication mechanisms, policy enforcement, and challenges for the implementation of Zero Trust Architecture are discussed.

The paper presents a general overview of the trends for the adoption of Zero Trust Architecture, along with the research gaps. However, the trust evaluation model is not discussed, nor is the performance compared.

2.6 Summary and Gaps Identified

The literature review reveals various aspects of the growth of Zero Trust Architecture from concept to reality in terms of enterprise deployments and theoretical trust evaluation mechanisms. The Zero Trust Architecture proposed by NIST is a standardized reference architecture that defines the key components and principles needed to implement Zero Trust systems. Google's BeyondCorp is another example of the viability of Zero Trust deployments in enterprise networks by removing implicit network trust and using identity and device verification mechanisms. Various survey-based research studies also discuss different ZTNA deployment frameworks and architectures, while research in trust and reputation mechanisms discusses mathematical formulations to calculate trust values using interactions in a system. Other recent literature review studies discuss various aspects of Zero Trust deployments in different domains, including cloud computing, IoT, healthcare, and enterprise networks.

Comparison

Key Dimension	**NIST SP 800-207**	**BeyondCorp**	**ZTNA Review**	**Trust Systems**	**ZTA Literature Review**
Trust Model	Continuous verification using identity, device posture and policies	Identity and device-based trust before application access	Compares multiple ZTNA trust evaluation approaches	Trust calculated using reputation and probabilistic models	Continuous verification using identity, device and contextual parameters
Access Control	Policy-based authorization using PDP and PEP	Identity-aware proxy enforcing per-application access	Broker-based and agent-based access models	Access decisions based on calculated trust scores	Policy-based access control with identity and device validation
Implementation Depth	High-level architectural reference model	Real-world enterprise deployment model	Comparative academic survey of frameworks	Theoretical trust computation models	Cross-domain review of ZTA implementations
Contribution	Defines fundamental Zero Trust principles	Demonstrates practical perimeterless security architecture	Identifies deployment trends and architectural gaps	Provides trust evaluation algorithms	Highlights adoption trends and research challenges

Table 2.1: Comparison of Existing Zero Trust Approaches

Research Gaps Identified

From the above literature review, the following gaps in the existing Zero Trust studies have been identified:

1. Most of the Zero Trust studies have only proposed the architecture of Zero Trust but have failed to include the implementation of the trust evaluation mechanisms.
2. Real-time trust score calculation involving multiple security events is seldom implemented in the existing systems.
3. Limited work has been done in the integration of behavioral anomaly detection mechanisms along with the Zero Trust policy engines.
4. Very few studies have proposed a complete end-to-end Zero Trust system.
5. There is a lack of dynamic access control mechanisms.

2.7 Summary of Chapter

This chapter has discussed the important research works related to the concept of Zero Trust Architecture. This includes the standard framework defined by the NIST organization, the BeyondCorp model defined by the Google organization, architectural surveys of the concept of ZTNA, as well as the concept of trust computation. This provides a base for the concept of the proposed Zero Trust Policy Engine.

Chapter 3

System Design

The system design phase describes the architectural components of the proposed Zero Trust Micro-VPN system. In this phase, the design of the system is based on the implementation of secure communication, authentication, and access control for the campus network. The system incorporates different security features such as VPN communication, multi-factor authentication, trust score evaluation, and microsegmented access to the campus network. In this design, all access requests are verified, authorized, and monitored before access is granted to sensitive resources.

3.1 Overall Architecture

The proposed system is designed to operate under the Zero Trust security model. In this model, none of the users or devices are inherently trusted. Each request is authenticated, authorized, and validated before allowing access to the system. The Zero Trust model consists of four major layers: authentication and access control, secure VPN communication, Zero Trust policy enforcement, and role-based campus portal services.

Firstly, the system is accessed through a secure VPN tunnel established between the client device and the server. The authentication layer is responsible for authenticating users through password-based authentication combined with multi-factor authentication (MFA) using Google Authenticator (TOTP). Once authentication is successful, a JWT token is generated that contains session information of the user.

Each request is validated through the Zero Trust policy engine. The policy engine checks role-based permissions, trust score values, and device fingerprint information before granting access. The system also operates through microsegmentation of different services such as student management, faculty management, and administrative control.

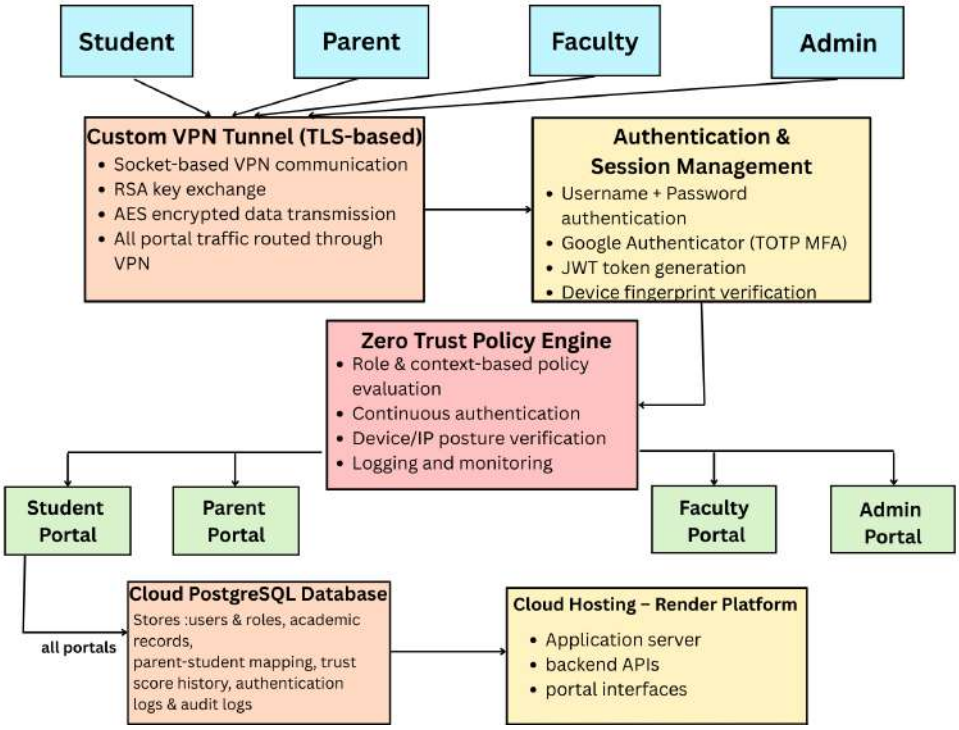


Figure 3.1: Overall Architecture of Zero Trust Micro VPN System

3.2 Component Design

The system consists of various interconnected components working together to achieve the concept of the Zero Trust security model. Some of the key components are authentication services, VPN communication modules, policy enforcement mechanisms, and role-based application portals.

The Authentication Component manages the user login process, password verification, and multi-factor authentication using Google Authenticator. It provides JWT tokens to authenticated users and handles secure session management. The Secure Communication Component provides an encrypted VPN tunnel using hybrid cryptography, with RSA key exchange and AES data encryption for efficient communication.

The Zero Trust Security Engine analyzes each request based on user identification, role permissions, device fingerprinting, and behavioral trust scores. The Campus Portal Component provides separate portals for students, parents, faculty, and administrators to restrict each role to the resources they are permitted to use.

3.3 Module Division

The proposed system is divided into four main modules to ensure modular design and easier implementation.

Module 1: Authentication and Access Control Layer

This module provides secure authentication for users using a combination of username, password, and multi-factor authentication through Google Authenticator (TOTP). JWT tokens are issued after successful authentication to provide secure user sessions. Role-based access control policies and trust score evaluation are integrated into this module.

Trust Score Evaluation: To provide adaptive access control, the system uses a Trust Score Evaluation mechanism for each user session. A dynamic trust score, denoted by the variable T, is assigned to each user based on the behavior of the user during the authentication process. The trust score begins with a maximum value and decreases based on security-related events observed during user interaction.

The events contributing to trust score reduction include failed login attempts, failed authentication attempts, RBAC policy violations, and rate limit violations. A penalty is assigned to each event, and the trust score is determined by subtracting the sum of penalty points from the maximum score.

$$T = 100 - (\alpha \times failed_logins) - (\beta \times failed_auth_attempts) - (\gamma \times rbac_violations) - (\delta \times rate_limit_hits)$$

Event	**Symbol**	**Weight**
Failed login	α	25
Failed authentication	β	20
RBAC violation	γ	15
Rate limit hit	δ	10

Table 3.1: Penalty Weights for Trust Score Calculation

Based on the trust score calculated, the system adapts the level of access provided to the user. If the trust score drops below a certain threshold, the system forces the user to re-authenticate or denies access to protected resources.

3.3.1 Zero Trust VPN Backend Architecture

The backend architecture for the proposed Zero Trust VPN system follows a layered encryption and authentication protocol to ensure secure communication between the client and server. Every request from the VPN follows a protocol with an encrypted payload containing the message length, encrypted AES session key with RSA-2048 encryption, initialization vector, and the encrypted data.

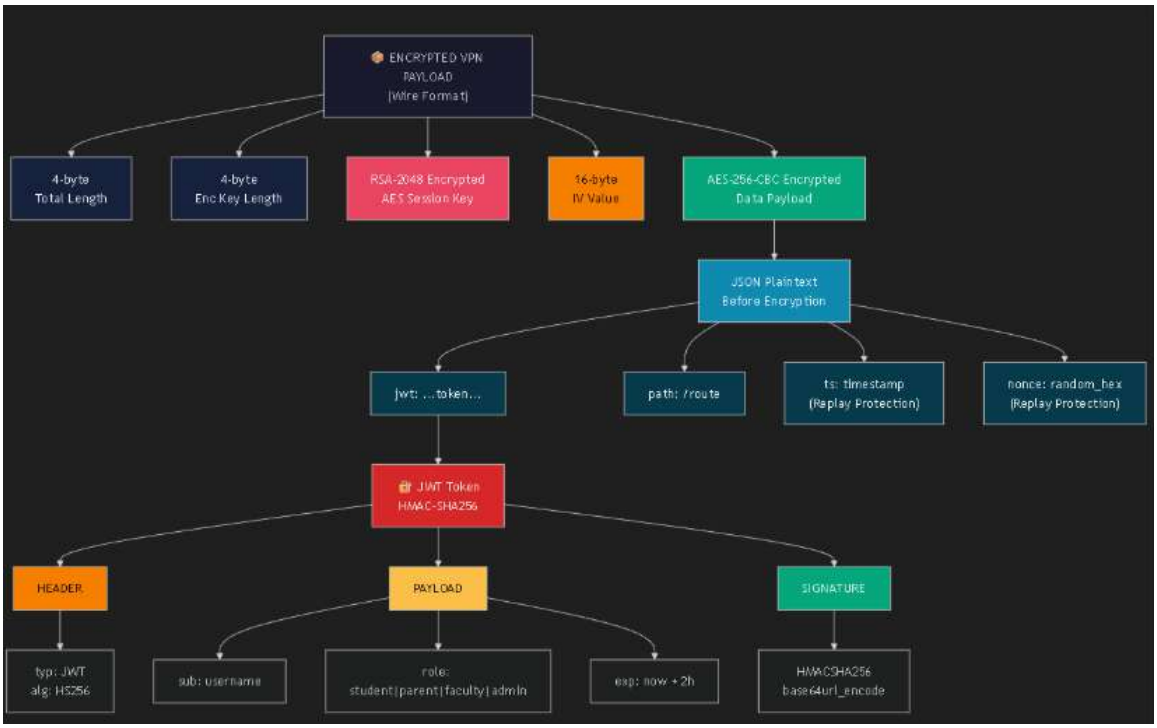


Figure 3.2: Zero Trust VPN Backend and Access Control Engine

Before encryption, the data is structured in JSON format containing the request route, timestamp to prevent replay attacks, and a random nonce value. Authentication uses JWT tokens with HMAC SHA-256 signatures containing user identity, role, and expiration time. This architecture ensures confidentiality, integrity, and secure access control.

Module 2: Secure VPN Tunnel and Cryptographic Communication Layer

This module establishes a custom socket-based VPN tunnel to provide secure communication between the client and the server. Hybrid encryption secures the data transmission process. RSA encryption is used for secure key exchange during communication, while AES encryption provides symmetric encryption of the data packets for efficient transmission.

A custom socket-based VPN tunnel facilitates encrypted client-server communication. A hybrid encryption scheme is implemented where RSA securely exchanges session keys while AES encrypts the transmitted data. All academic portal communication occurs strictly through the encrypted VPN tunnel. Secure session creation and controlled tunnel teardown mechanisms ensure safe communication.

Cryptographic logging using AES-GCM is implemented to securely store VPN session logs with authenticated encryption.

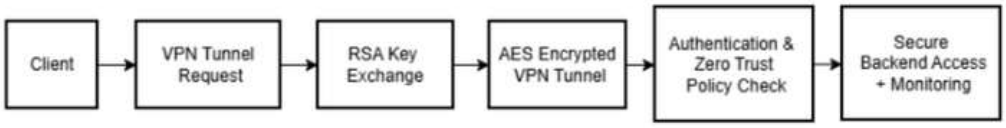


Figure 3.3: Secure VPN Tunnel and Cryptographic Communication Layer

During session establishment, RSA encrypts the symmetric AES session key exchanged between client and server. The server sends its public key, which the client uses to encrypt the session key. Only the server can decrypt this key, ensuring secure communication.

Once exchanged, AES symmetric encryption is used for all further communication through the VPN tunnel, ensuring confidentiality and preventing unauthorized access.

Module 3: Zero Trust Security Engine

This module enforces the core security mechanisms of the Zero Trust model. Unlike traditional security models, the Zero Trust system continuously verifies user identity during interactions with the system. Least privilege access control is implemented using Role-Based Access Control (RBAC).

A dynamic trust score system continuously evaluates session trustworthiness based on events such as login failures, authentication failures, role violations, and abnormal access behavior. The system also implements device fingerprinting by combining the user's IP address with User-Agent information to detect device changes during system access.

Session management features such as inactivity timeouts and restrictions on concurrent sessions prevent misuse of sessions. If the trust score falls below certain thresholds, adaptive responses are triggered to protect the system.

During security incidents, the system can activate Emergency Read-Only Mode, temporarily blocking all write operations while allowing monitoring access. All authentication

events, trust score changes, and policy violations are logged through centralized monitoring mechanisms.

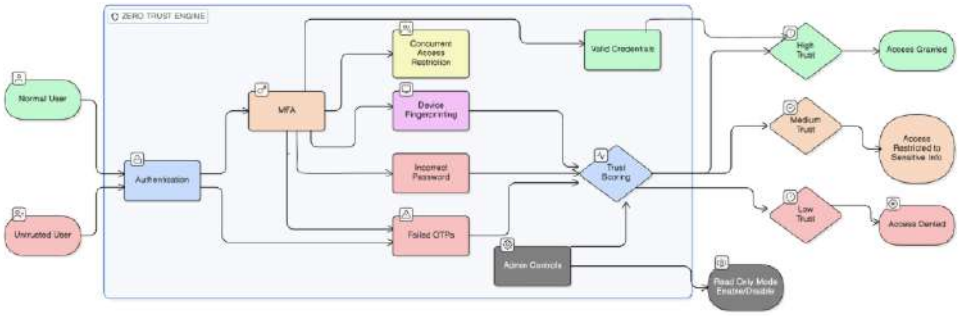


Figure 3.4: Architecture of Zero Trust Security Engine

Module 4: Secure Role-Based Campus Portals

This module includes portals for different academic roles, ensuring secure handling of institutional services while enforcing strict access control policies. Students, Parents, Faculty, and Administrators are provided with dedicated portals aligned with their responsibilities.

The Student Portal allows students to securely view marks, attendance records, assignments, notices, and submit grievances.

The Parent Portal provides read-only monitoring of their child's academic information including marks, attendance, and announcements.

The Faculty Portal allows faculty members to manage academic records such as marks entry, attendance tracking, assignment publishing, and academic notices.

The Administrator Portal provides system-level controls including class creation, student enrollment, faculty assignment, batch-based roll number generation, and linking parents to students through email mapping. Administrative operations are strictly monitored and logged.

All portals operate within the secure VPN environment, ensuring encrypted communication and strict validation of every request through the Zero Trust security engine.

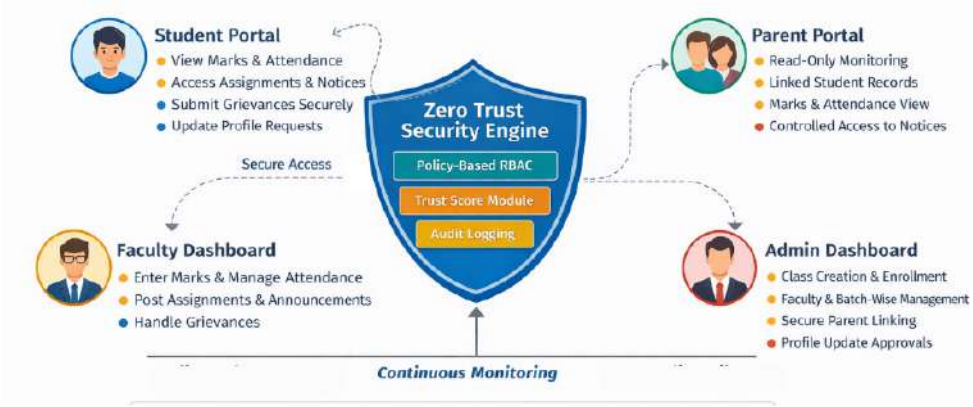


Figure 3.5: Architecture of Secure Role-Based Campus Portals

3.4 Tools and Technologies

The system is implemented using multiple software tools and infrastructure technologies.

- **Software Technologies**

  Python with Flask is used for backend development. PostgreSQL is used as the database to store user credentials, academic data, and system logs. HTML, CSS, and Bootstrap are used to build responsive user interfaces. Cryptographic libraries implement secure communication using RSA and AES encryption techniques. JWT libraries provide token-based authentication and session management. GitHub is used for version control, and the system is deployed on the Render cloud platform for hosting and testing.

- **Hardware Requirements**

  A laptop or desktop system with a minimum of 8 GB RAM is required for development and deployment. The system requires a stable internet connection for authentication, secure communication, and cloud-based deployment. The application can be deployed on local servers or cloud infrastructure within a campus network environment.

3.5 Work Breakdown and Responsibilities

The project was implemented collaboratively by the team.

- **Alan Suresh:** Designed and implemented the Secure VPN Tunnel including tunnel establishment, connection management, authentication integration, encrypted communication flow, cryptographic security logging and VPN activity logging.
- **Aparna V Sunil:** Developed the Secure VPN cryptographic layer including RSA-based key exchange, AES encryption and decryption mechanisms, session key management, data integrity validation, authentication integration using Google Authenticator.
- **Abhirami C U:** Designed and implemented the Zero Trust Policy Engine including trust score evaluation, device fingerprinting, session restriction, dynamic access control, and centralized audit logging.
- **Aryasree Nambiar:** Designed and developed the Student, Parent, Faculty, and Admin portals including role-based access control, academic management features, session timeout mechanisms, and audit logging.

3.6 Key Deliverables

Key deliverables of the project include the implementation of the secure Zero Trust Micro-VPN architecture, the Zero Trust policy engine, encrypted VPN communication modules, and the role-based campus academic management portals. Additional deliverables include system documentation, design diagrams, and testing results validating the effectiveness of the proposed security architecture.

3.7 Project Timeline

The project was developed following a structured timeline consisting of problem analysis, literature study, system design, implementation, testing, and documentation phases.

Figure 3.6: Project Timeline (Gantt Chart)

3.8 Summary of chapter

In this chapter, the system design of the proposed Zero Trust Micro-VPN architecture for campus networks has been discussed. The overall architecture, system components, module structure, and technology stack used in the implementation were explained. The roles of each team member and the project development timeline were also described. This design forms the foundation for implementing the secure Zero Trust academic management system described in the next chapter.

Chapter 4

Results and Discussions

This section presents the results of implementing the proposed Zero Trust Micro-VPN system for secure campus network access. The system was tested in a controlled development environment to assess its functionality, security measures, and access control methods. Key components like multi-factor authentication, role-based access control, trust score evaluation, and secure session management were verified through various testing scenarios. The results show how the system effectively limits unauthorized access, enforces strict role separation, and continuously evaluates user behavior to ensure secure operations across the campus portals.

4.1 System Deployment

The proposed Zero Trust Micro-VPN system was successfully deployed on the cloud platform using Render. The backend application developed using Python Flask was hosted on the cloud server, while PostgreSQL was used for secure storage of user credentials, academic records, and system logs. The deployment enables secure access to the campus portal through encrypted VPN communication and multi-factor authentication using Google Authenticator.

4.2 User Authentication Process

The authentication system was tested using multiple user roles including students, faculty, parents, and administrators. The login process requires a valid username and password followed by multi-factor authentication using Google Authenticator (TOTP).

Once authentication is successful, the system generates a JWT session token and allows access based on role-based access control policies.

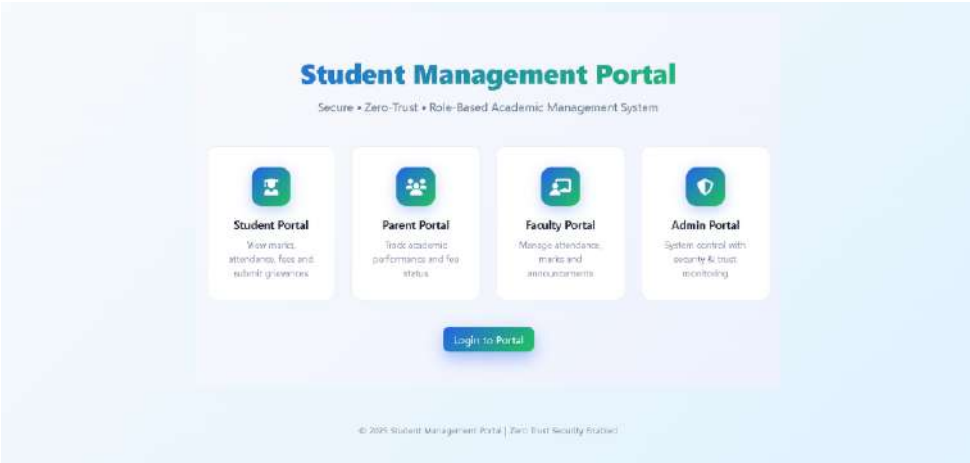


Figure 4.1: Login Page

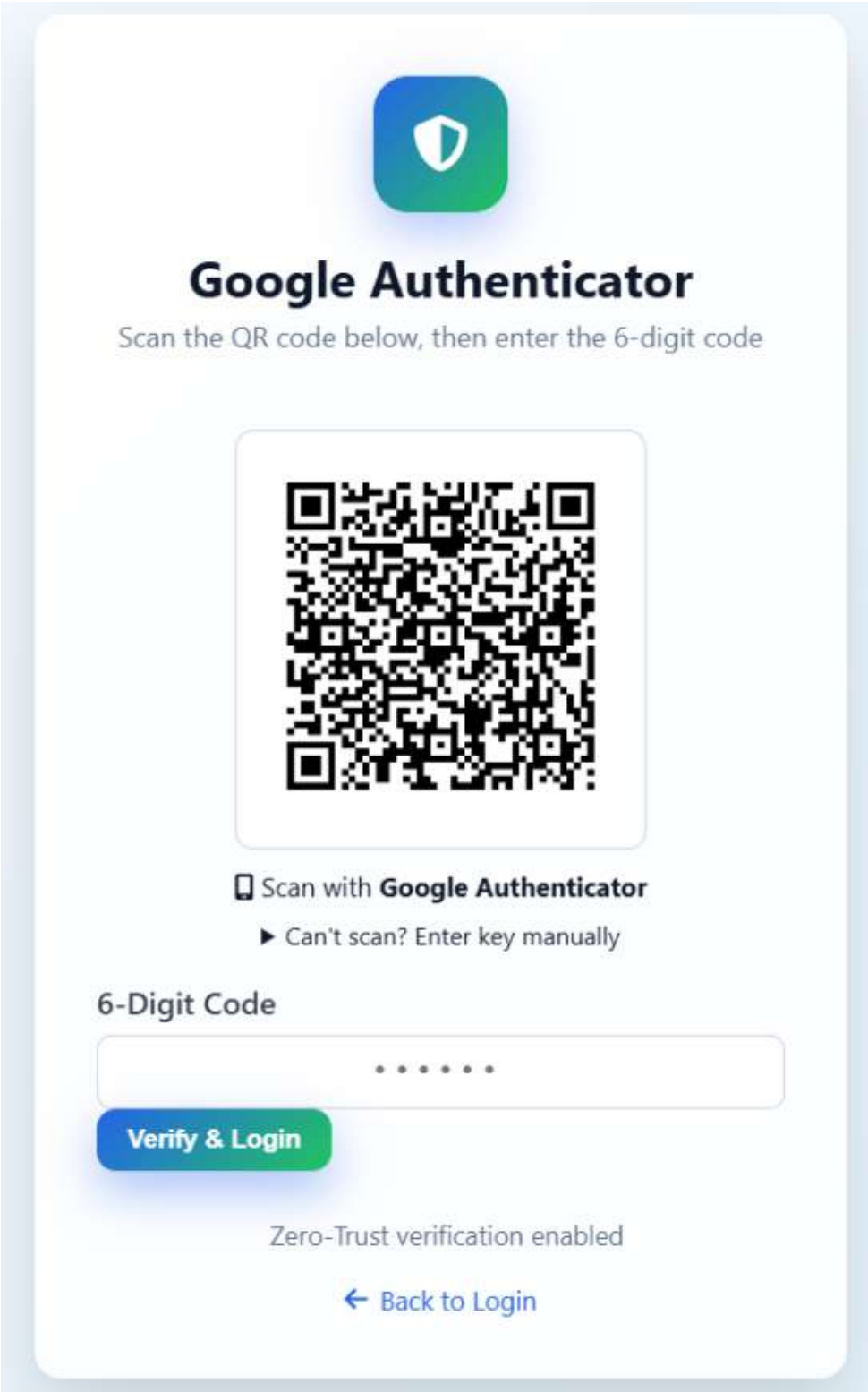


Figure 4.2: Google Authenticator

4.3 VPN Tunnel Establishment

The system establishes a secure VPN tunnel between the client and the server using a custom socket-based implementation. RSA encryption is used for secure key exchange while AES encryption ensures confidentiality of transmitted data.

All communication between the client application and campus portal occurs through the encrypted VPN tunnel, preventing unauthorized access or interception of sensitive data.

4.4 Role-Based Portal Access

The system successfully enforces role-based access control by providing separate portals for students, parents, faculty members, and administrators.

Each user role has access only to the resources permitted by the Zero Trust policy engine. For example, students can view academic records while faculty members can update marks and attendance.

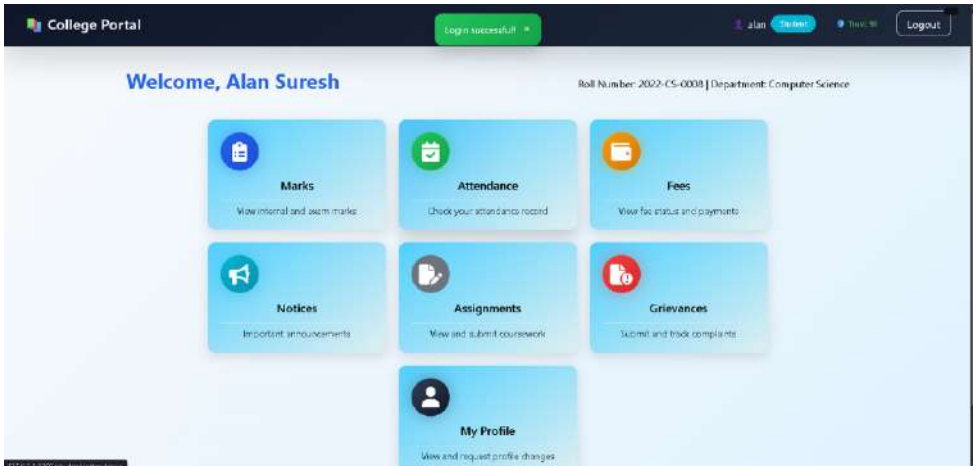


Figure 4.3: Student Dashboard

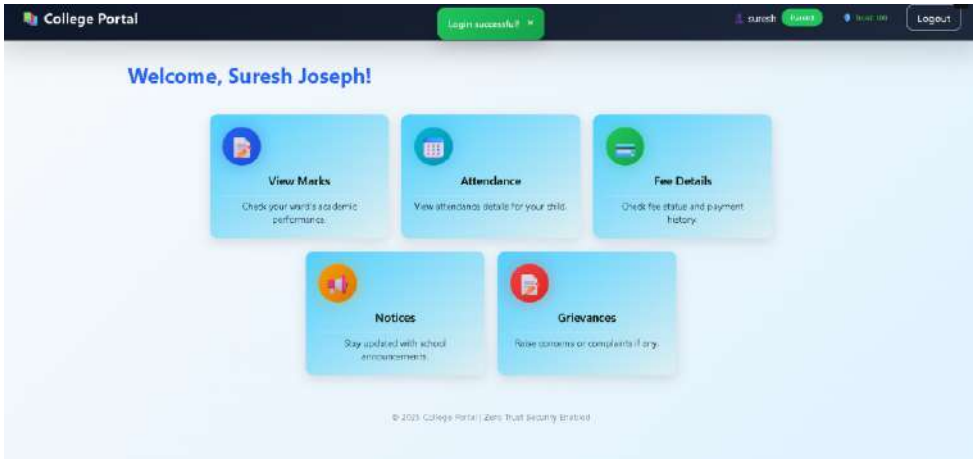


Figure 4.4: Parent Dashboard

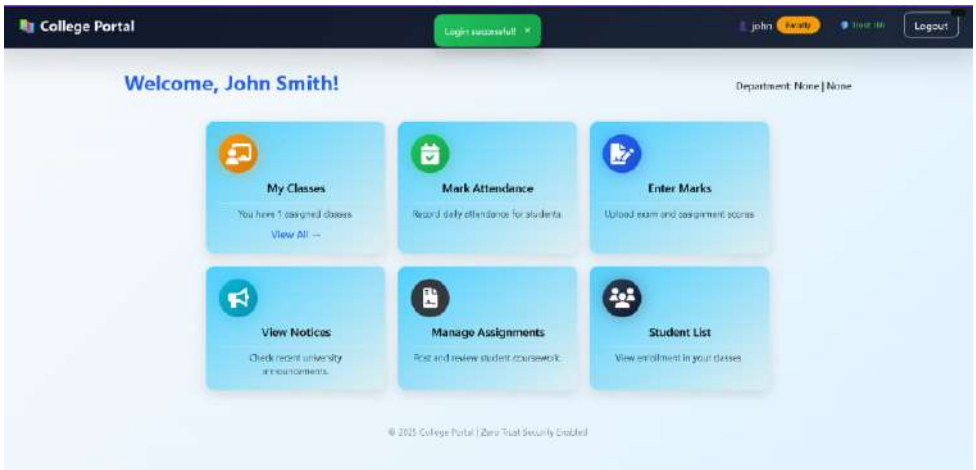


Figure 4.5: Faculty Dashboard

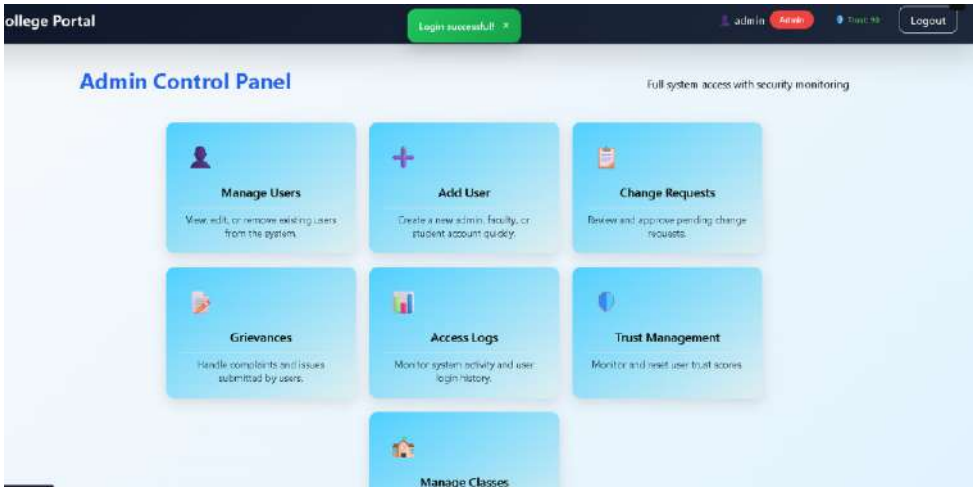


Figure 4.6: Admin Dashboard

4.5 Trust Score Evaluation

The Zero Trust policy engine dynamically evaluates user behavior through a trust score mechanism. The trust score decreases when suspicious events such as failed login attempts, RBAC violations, or abnormal requests are detected.

If the trust score falls below the defined threshold, the system restricts access and forces re-authentication to ensure security.

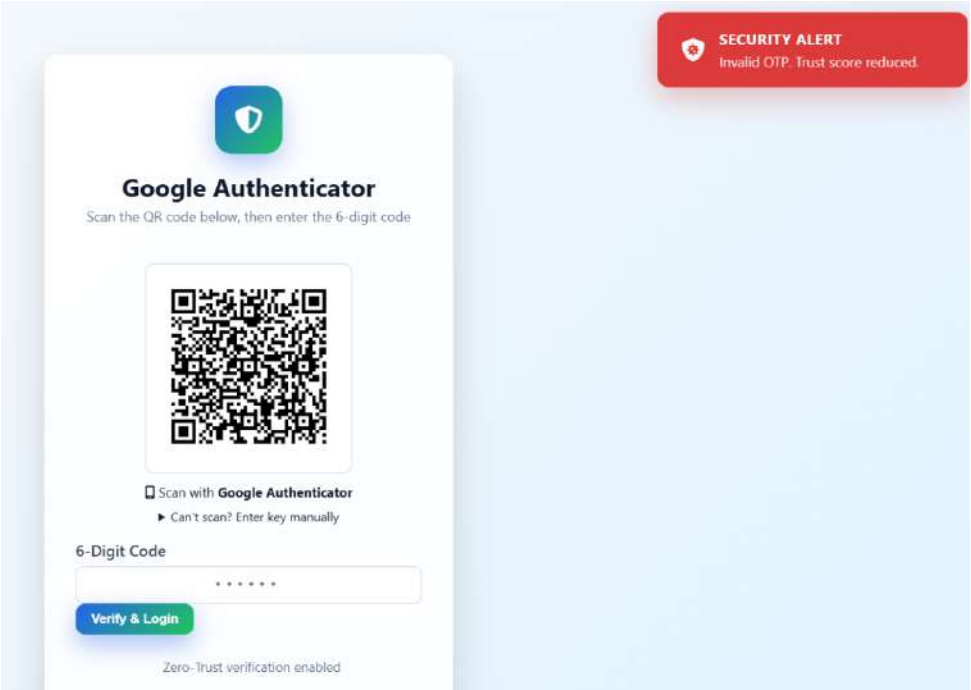


Figure 4.7: Trust Score Reduction Due To Invalid OTP

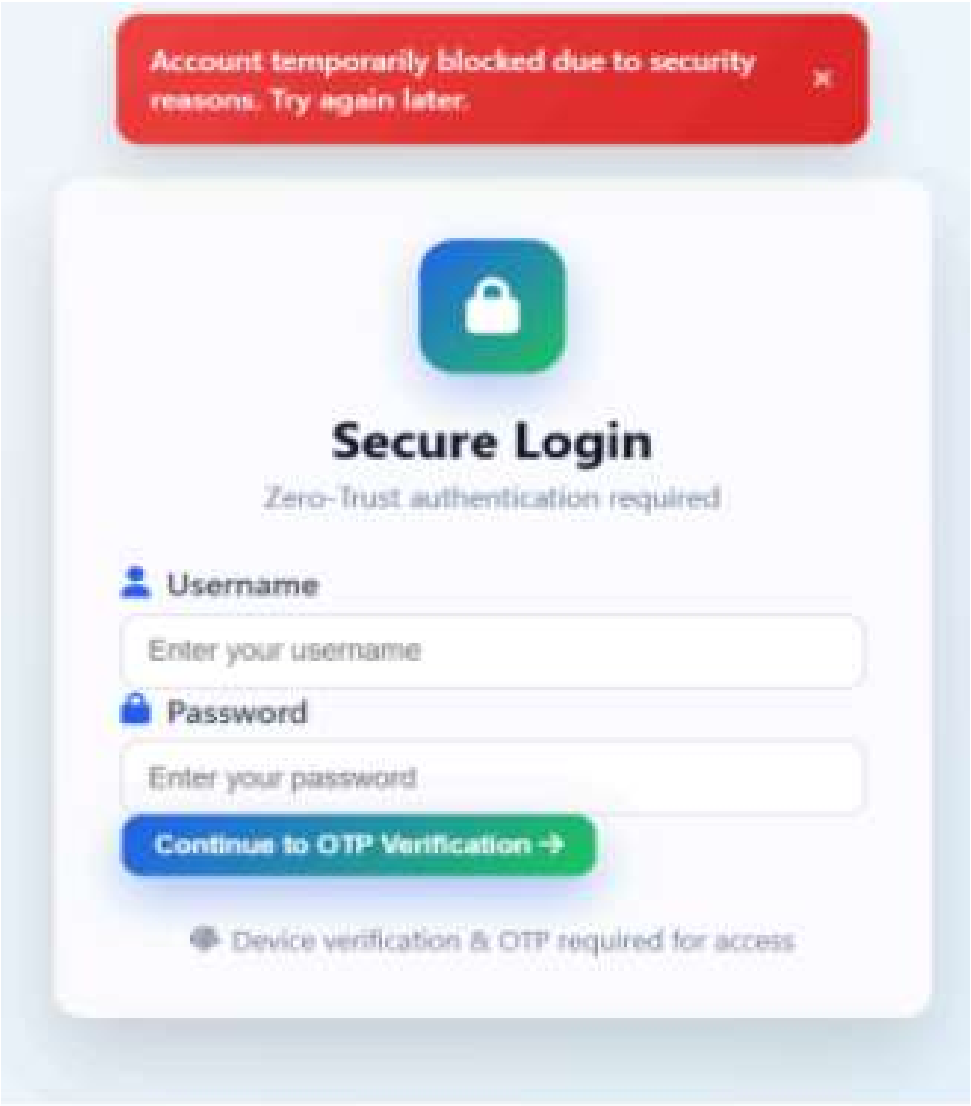


Figure 4.8: Account Temporarily Blocked Due To Very Low Trust Score

Figure 4.9: Limited Access Due To Low Trust Score

Chapter 5

Conclusions & Future Scope

5.1 Conclusion

This chapter summarizes the outcomes of the project "Design and Implementation of a Zero Trust Micro-VPN for Campus Networks." The primary goal of this project was to design and implement a secure campus network architecture that integrates Zero Trust principles with a custom Micro-VPN system and a role-based academic management portal. Through this project, the team successfully demonstrated how modern security mechanisms such as continuous authentication, encrypted communication, and strict access control can be applied to protect sensitive institutional resources.

The project involved the design and implementation of a secure VPN tunnel with hybrid encryption using RSA for key exchange and AES for data encryption. Authentication and session management were strengthened using multi-factor authentication through Google Authenticator and JWT-based token validation. The system also incorporated a Zero Trust policy engine that evaluates user identity, role permissions, trust score, and device fingerprinting before granting access to protected resources. A role-based campus portal was developed for students, parents, faculty, and administrators to securely manage academic activities such as marks, attendance, assignments, announcements, and grievance handling. The system uses a PostgreSQL database for secure storage of academic and authentication data and has been deployed on a cloud platform to demonstrate real-world accessibility and scalability.

5.2 Future Scope

In the future, the system can be extended by integrating advanced behavioral analytics and machine learning models to enhance trust score evaluation and anomaly detection. Additional security features such as biometric authentication, adaptive risk-based access

control, and integration with institutional identity management systems can also be incorporated. Furthermore, the architecture can be expanded to support large-scale enterprise or multi-campus deployments with improved scalability, distributed VPN gateways, and enhanced monitoring tools for real-time security analysis.

References

[1] S. Rose, O. Borchert, S. Mitchell, and S. Connelly, "Zero Trust Architecture," NIST Special Publication 800-207, National Institute of Standards and Technology, 2020.

[2] R. Ward and B. Beyer, "BeyondCorp: A New Approach to Enterprise Security," ;login: The USENIX Magazine, vol. 39, no. 6, pp. 6–11, 2014.

[3] A. Kumar et al., "A Comprehensive Review of Zero Trust Network Architecture (ZTNA) and Deployment Frameworks," IEEE Access, vol. 9, 2021.

[4] A. Jøsang, R. Ismail, and C. Boyd, "A Survey of Trust and Reputation Systems for Online Service Provision," Decision Support Systems, vol. 43, no. 2, pp. 618–644, 2007.

[5] M. L. Gambo and A. Almulhem, "A Systematic Literature Review on the Implementation and Challenges of Zero Trust Architecture Across Domains," IEEE Access, vol. 13, 2025.

Appendix A: Presentation

100% Project Presentation

Group 6

DESIGN AND IMPLEMENTATION OF A ZERO TRUST MICRO-VPN FOR CAMPUS NETWORKS

Guide: Mr. Sandy Joseph

Team Members:

- Alan Suresh (U2203023)
- Aparna V Sunil (U2203051)
- Abhirami C U (U2203010)
- Aryasree Nambiar (U2203054)

Problem Definition

Traditional campus systems trust users after login, making them vulnerable to insider threats and unauthorized access. With increased remote access and personal devices, a Zero Trust–based model is required to ensure continuous verification and secure, role-based access to campus resources

What Is Zero Trust?

❝ Never trust. Always verify. ❞

Instead of trusting users just because they are inside the network (like traditional VPNs), every request is:

- Authenticated
- Authorized
- Continuously validated
- Logged
- Risk evaluated

Objectives

- To develop a secure VPN tunnel for safe communication within the campus network.
- To implement robust key management and authentication for all users.
- To provide secure role-based access for Student, Parent, Faculty, and Admin portals.
- To enforce session validation, auditing, and access logging for all users.
- To ensure that all student management portal data flows only through authenticated and authorized channels.

Purpose and Need

Purpose

- Implement a Zero Trust–based Micro VPN for secure campus network access
- Enforce identity-based and role-based microsegmented access control
- Introduce continuous authentication and adaptive trust scoring for enhanced security

Need

- Traditional VPNs provide broad network access after login, creating security vulnerabilities
- Sensitive campus data (marks, attendance, fees, personal records) requires strong protection
- Need to prevent unauthorized access, privilege escalation, and lateral movement

Literature Survey – Paper I

S. Rose et al., NIST SP 800-207, 2020 [1]

Methodology – The paper defines the architectural foundation of Zero Trust by introducing Policy Decision Point (PDP), Policy Enforcement Point (PEP), and continuous verification principles. It shifts security from perimeter-based protection to identity-, device-, and context-aware access control.

Advantages –

- Industry Standard Framework
- Clear Architectural Components (PDP/PEP)
- Continuous Authentication Model
- Scalable for Enterprise Networks

Disadvantages –

- Provides high-level guidance without implementation-level detail
- Does not address computational overhead of continuous verification
- Requires mature identity and asset inventory systems
- Transition from legacy perimeter-based networks is complex

Literature Survey – Paper 2

BeyondCorp: A New Approach to Enterprise Security [7]

Methodology – Introduces Google's real-world Zero Trust implementation that enforces per-request authentication and device-based access control instead of trusting internal networks.

Advantages –

- Real-World Deployment Model
- Identity-Centric Access Control
- Device Posture Verification
- Eliminates Network Perimeter Trust

Disadvantages –

- Requires extensive infrastructure automation
- High operational cost for device posture validation
- Complex certificate and identity lifecycle management
- May introduce latency due to per-request evaluation

Literature Survey – Paper 3

A Comprehensive Review of Zero Trust Network Architecture (ZTNA) and Deployment Frameworks [2]

Methodology – Provides a detailed survey of Zero Trust architectures, deployment strategies, policy engines, and trust evaluation mechanisms across enterprise and cloud environments.

Advantages –

- Comprehensive Architectural Overview
- Discusses Trust Evaluation Approaches
- Covers Deployment Challenges
- Useful for Implementation Planning

Disadvantages –

- Primarily theoretical and survey-based
- Lacks experimental implementation results
- Does not benchmark performance trade-offs
- Limited discussion on behavioral anomaly detection models

Literature Survey – Paper 4

A Survey of Trust and Reputation Systems for Online Service Provision [10]

Methodology – The paper introduces numerical trust modeling using weighted aggregation of positive and negative interactions. Trust is dynamically updated based on accumulated behavioral evidence and interaction history.

Advantages –

- Mathematical Trust Modeling
- Weighted Event-Based Trust Updates
- Dynamic Trust Recalculation
- Strong Theoretical Foundation

Disadvantages –

- Originally designed for reputation systems, not network access control
- Does not account for real-time session-level risk evaluation
- Assumes availability of large behavioral datasets
- Does not integrate directly with policy enforcement engines

Literature Survey – Paper 5

M. L. Gambo and A. Almulhem, "A Systematic Literature Review on the Implementation and Challenges of Zero Trust Architecture Across Domains," Sensors, 2025 [5]

Methodology – The study conducts a systematic literature review of Zero Trust Architecture implementations across sectors such as healthcare, cloud computing, IoT, and enterprise networks. It analyzes security models, authentication mechanisms, policy enforcement methods, and implementation challenges.

Advantages –

- Provides broad coverage of Zero Trust applications across domains
- Identifies common architectural components and design principles
- Highlights key challenges in ZTA adoption
- Useful for understanding trends and research gaps

Disadvantages –

- Primarily a survey paper without experimental implementation
- Does not propose a specific trust evaluation mechanism
- Limited discussion on real-time access decision models
- Lack of quantitative performance comparison

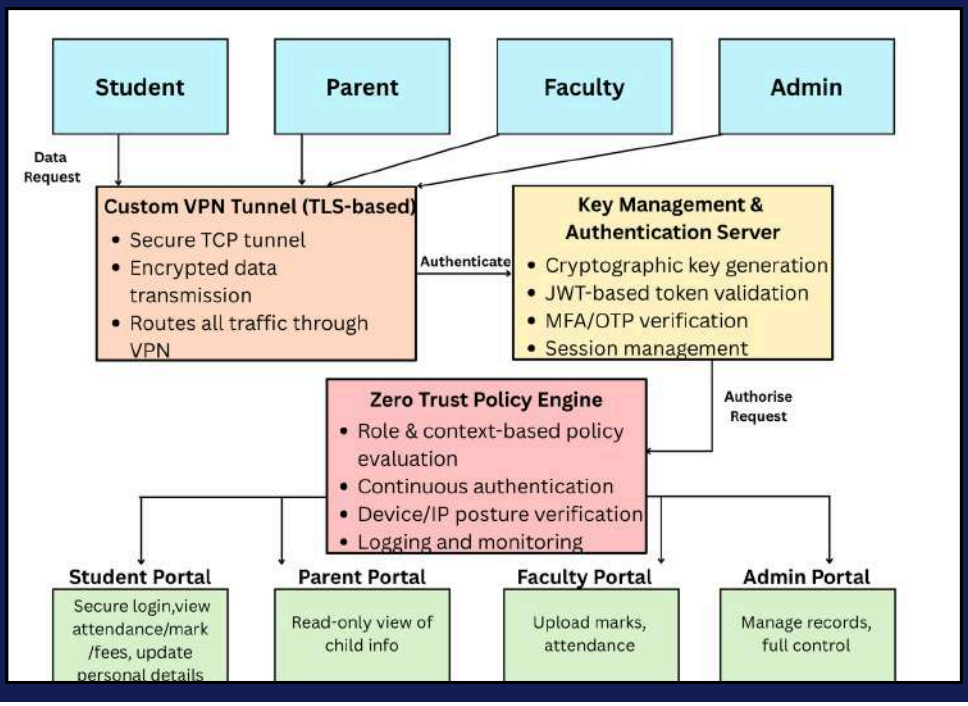
Comparison Table

Key Dimension	NIST SP 800-207	BeyondCorp: A New Approach to Enterprise Security	A Comprehensive Review of Zero Trust Network Architecture (ZTNA) and Deployment Frameworks	A Survey of Trust and Reputation Systems for Online Service Provision	"A Systematic Literature Review on the Implementation and Challenges of Zero Trust Architecture Across Domains,"
Trust Model	Eliminates implicit trust; continuous verification using identity, device posture, and policy	Strong identity + device-based trust before app access	Compares multiple ZTNA trust evaluation approaches (static, adaptive, risk-based)	Trust computed mathematically via reputation, feedback, probabilistic models	Continuous verification using identity, device, and contextual factors
Access Control Mechanism	Dynamic policy-based authorization using PDP and PEP components	Identity-aware proxy enforcing per-application access	Surveys broker-based, agent-based, SDP, and microsegmentation models	Access decisions based on calculated trust scores	Policy-based access control with identity and device validation
Practical Implementation Depth	High-level reference architecture (technology-agnostic guidance)	Real-world operational deployment at scale	Comparative academic synthesis of implementations	Theoretical frameworks and algorithms for trust computation	Review of ZTA implementations across cloud, IoT, healthcare, and enterprise systems
Contribution to Zero Trust Evolution	Standardizes core ZTA components and principles	First large-scale proof that perimeter-less security works	Identifies trends, gaps, and deployment challenges in ZTNA	Provides foundational trust models that could enhance ZTA policy engines	Identifies adoption trends, challenges, and research gaps in ZTA

Proposed Method

- Develop a custom secure VPN tunnel using socket-based communication
- Implement RSA-based key exchange and AES encryption for secure data transmission
- Integrate password + OTP-based Multi-Factor Authentication (MFA)
- Issue short-lived JWT sessions for authenticated users
- Design a Zero Trust Policy Engine for request validation
- Enforce strict Role-Based Access Control (RBAC) with microsegmentation
- Implement dynamic Trust Score evaluation based on user behavior
- Enable continuous session validation and automatic re-authentication
- Log authentication attempts, policy decisions, and security events
- Build a segmented student management portal to demonstrate real-world Zero Trust enforcement

Architecture diagram



Module 1: Authentication & Access Control Layer

- Implemented JWT-based authentication for secure, stateless session management.
- Integrated OTP-based Multi-Factor Authentication (MFA) with retry limits, failure tracking, and automated account lock mechanisms.
- Enforced continuous authentication through session validation and token expiry monitoring.
- Applied policy-based request validation to enforce strict Role-Based Access Control (RBAC).
- Designed a dynamic trust score evaluation system (0–100 scale) to assess behavioral risk.
- Triggered adaptive re-authentication when trust score falls below defined thresholds.

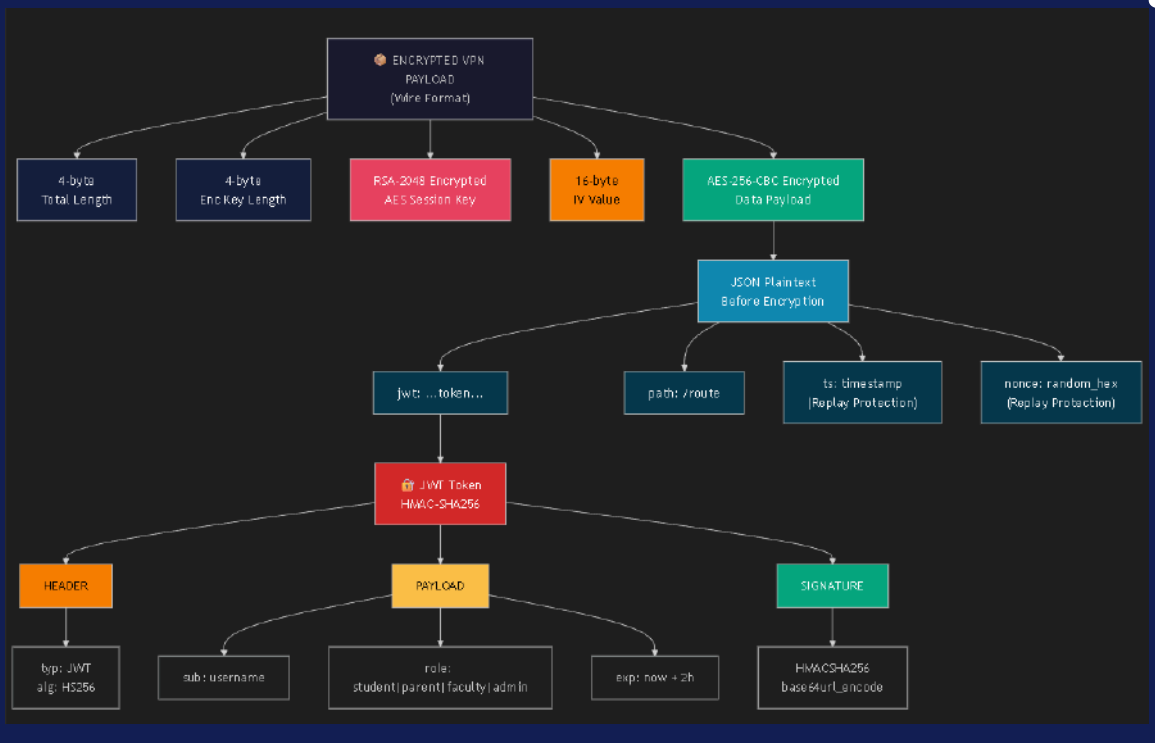
- Implemented security logging for authentication attempts, OTP events, policy decisions, and session activity.
- Conducted testing for session expiry handling, OTP retry limits, trust score updates, and JWT token re-issuance flows.

To compute Trust Score T,

```
T = 100
    - (α × failed_logins)
    - (β × failed_otps)
    - (γ × rbac_violations)
    - (δ × rate_limit_hits)
```

Event	Symbol	Weight
Failed login	α	25
Failed OTP	β	20
RBAC violation	γ	15
Rate limit hit	δ	10

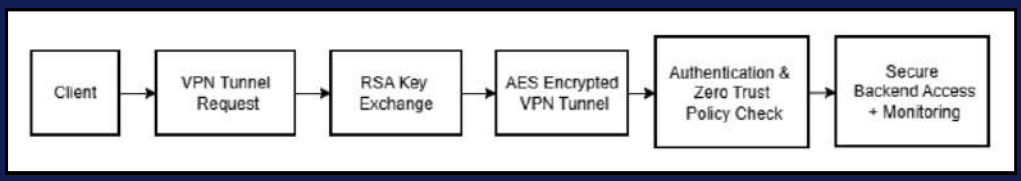
Zero Trust VPN Backend & Access Control Engine



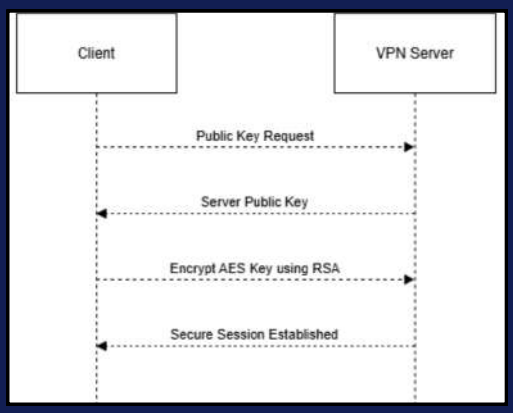
Module 2: Secure VPN Tunnel & Cryptographic Communication Layer

- Developed a custom socket-based VPN tunnel for encrypted client–server communication.
- Implemented Hybrid Encryption: RSA for secure session key exchange, AES for efficient symmetric data encryption.
- Ensured all academic portal traffic flows exclusively through the encrypted VPN tunnel.
- Managed secure session establishment and controlled tunnel teardown.
- Integrated with the Zero Trust authentication module to verify session tokens before allowing tunnel access.
- Implemented cryptographic logging using AES-GCM, ensuring VPN session logs are encrypted with authenticated encryption to maintain confidentiality, integrity, and tamper detection.

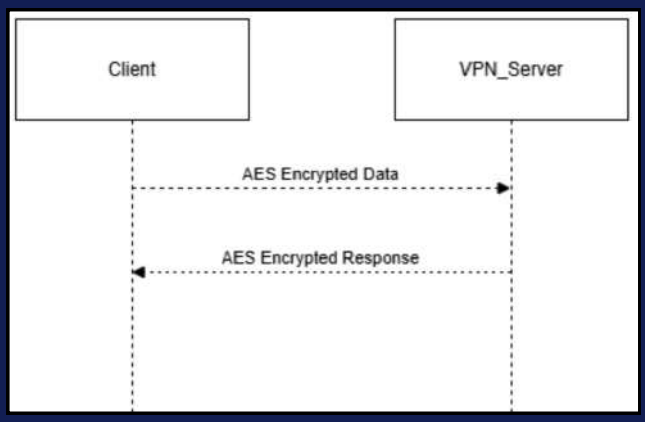
Secure VPN Tunnel & Cryptographic Layer



RSA – Secure Key Exchange



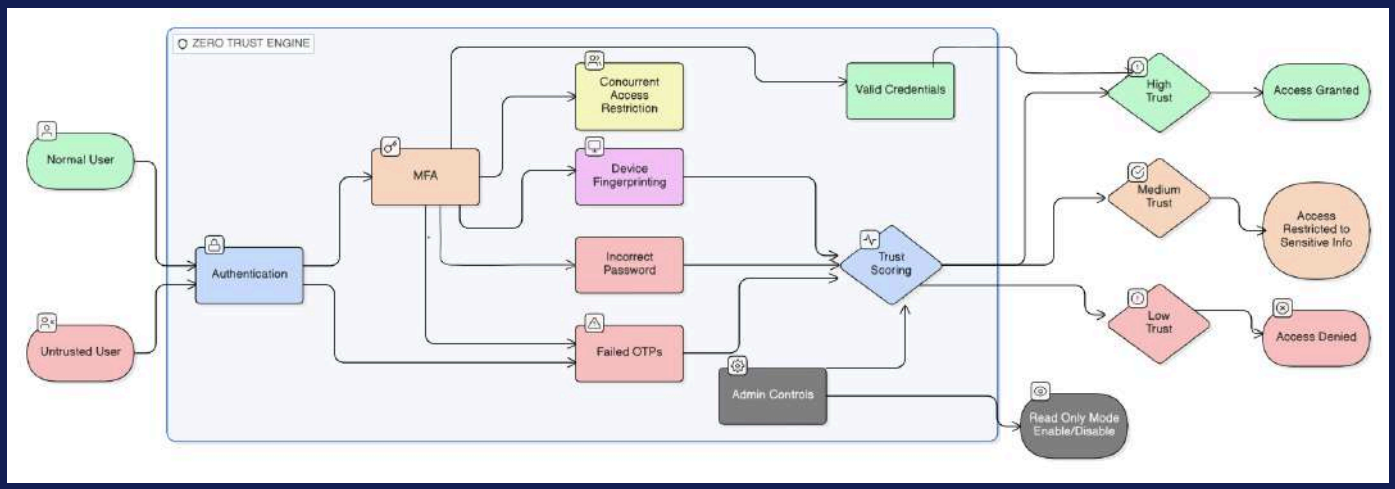
AES – Encrypted Data Transmission



Module 3: Zero Trust Security Engine

- Continuous verification to maintain user identity validation and session integrity
- Least-privilege access control using Role-Based Access Control (RBAC) and policy validation
- Dynamic Trust Score system updated based on login failures, OTP errors, role violations, and abnormal activity
- Device fingerprinting (IP address + User-Agent) to detect session hijacking and device switching
- Microsegmentation policies to isolate services and restrict lateral movement
- Session timeout and concurrent access restriction to prevent inactive or multiple simultaneous sessions
- Adaptive security responses such as re-authentication, restricted access, or access denial when trust thresholds are violated
- Emergency Read-Only Mode to block write operations during detected security incidents
- Centralized security logging and monitoring for authentication events, policy violations, and trust score updates

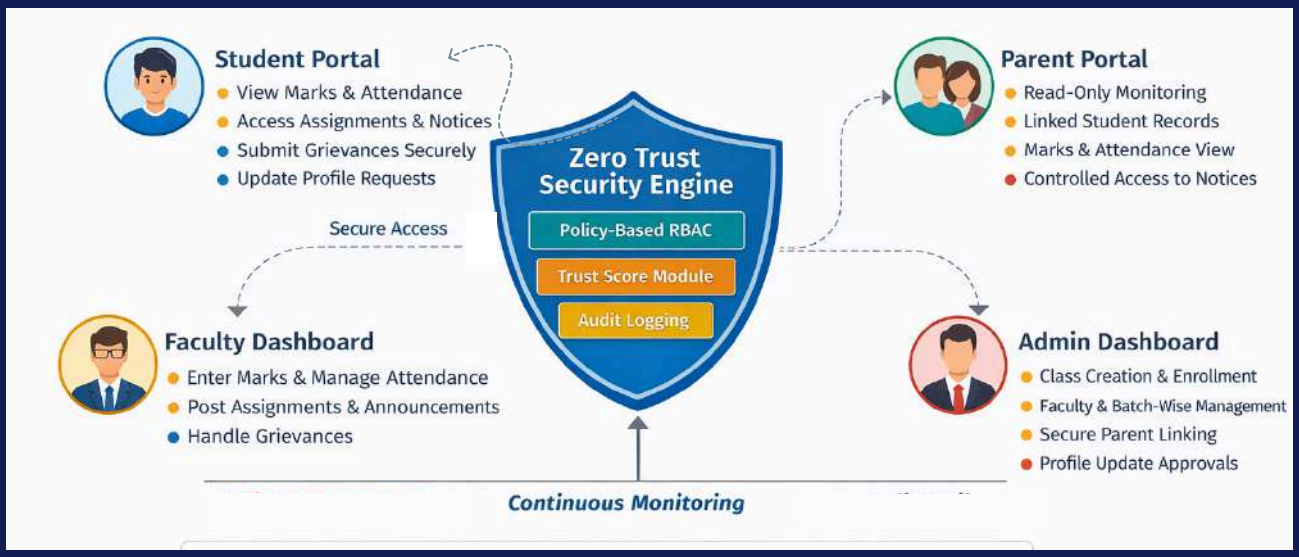
Architecture of Zero Trust Security Engine



Module 4: Secure Role-Based Campus Portals

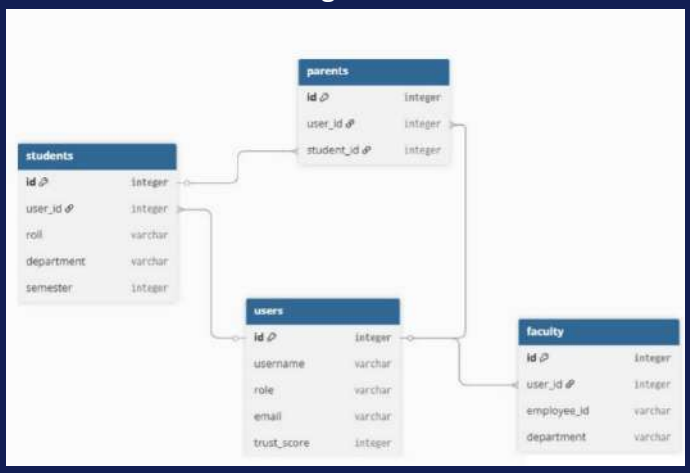
- Designed separate role-based portals for Students, Parents, Faculty, and Administrators with strict access boundaries.
- Enforced data isolation policies to ensure users can only access information permitted by their assigned roles.
- Implemented student features including marks viewing, attendance tracking, assignment access, notice boards, and grievance submission.
- Provided parents with secure read-only access to linked student academic records.
- Developed faculty modules for marks entry, attendance management, assignment distribution, announcements, and grievance resolution with controlled write permissions.
- Implemented administrative tools for class creation, student enrollment, faculty assignment, roll number generation, and parent–student linking.
- Integrated secure profile management, trust validation, and activity auditing to monitor portal interactions.
- Applied controlled database transaction mechanisms to prevent conflicting updates and ensure data integrity.

Architecture of Secure Role-Based Campus Portals

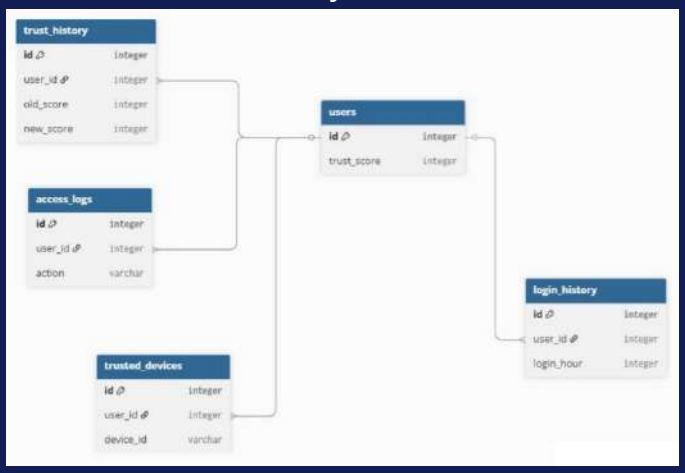


Database

User Management



Security and Audit

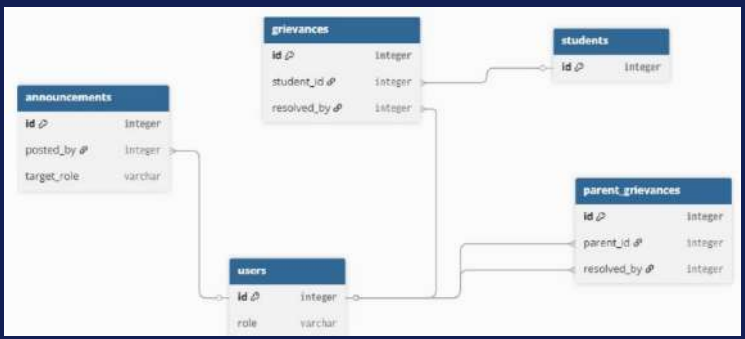


Database

Academic Records

Communication Handling

Assignment Processing



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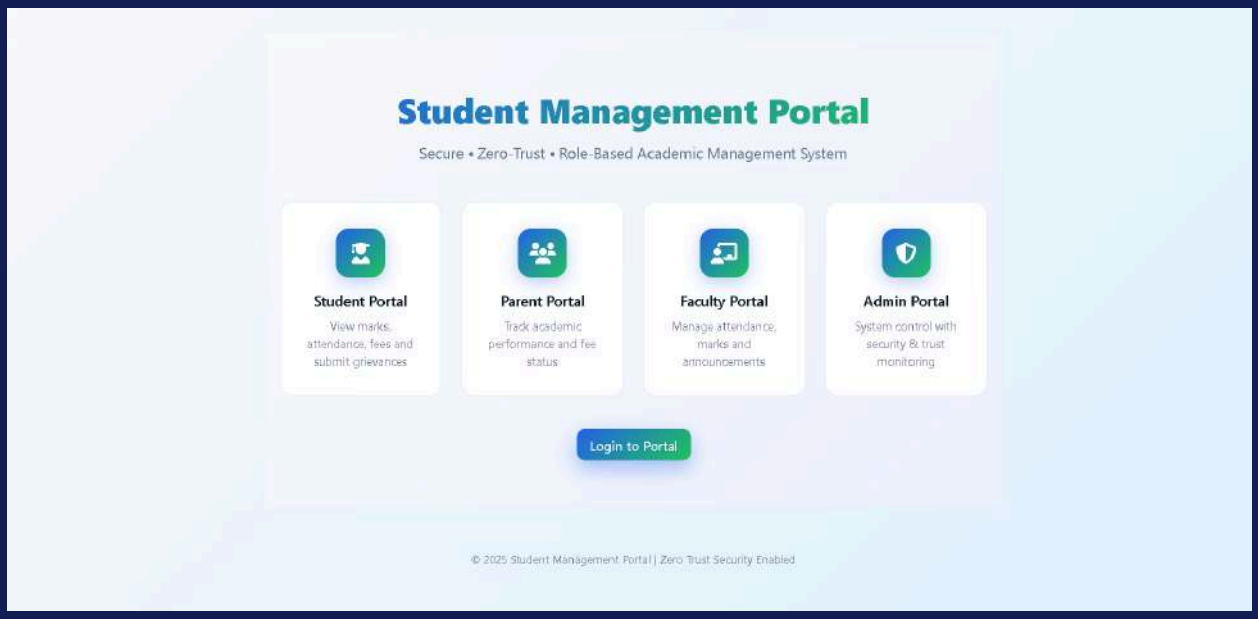

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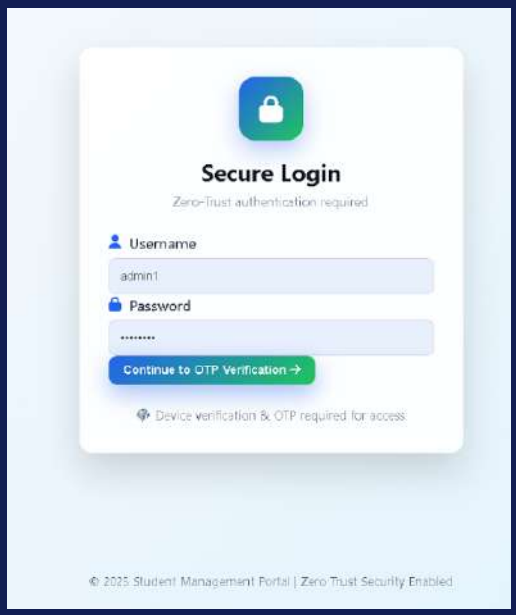

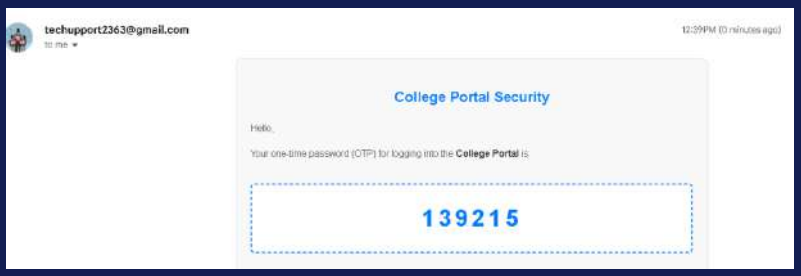

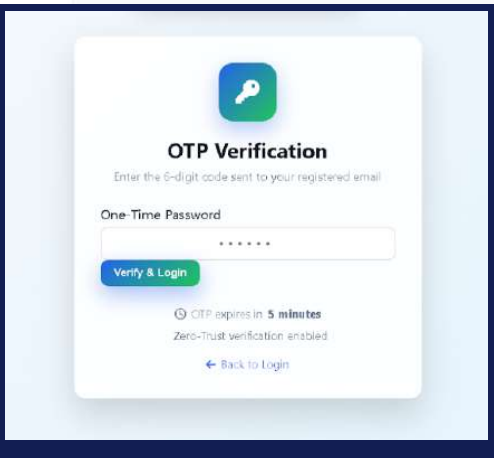

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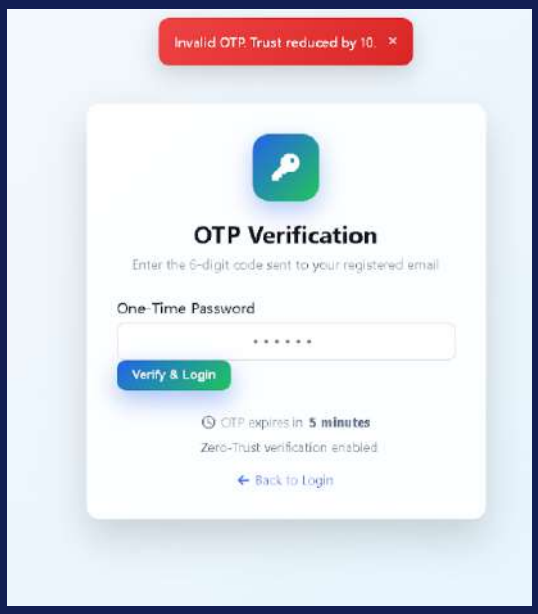

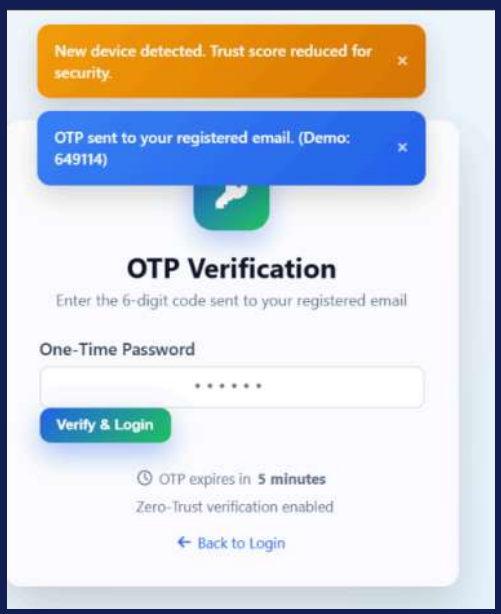

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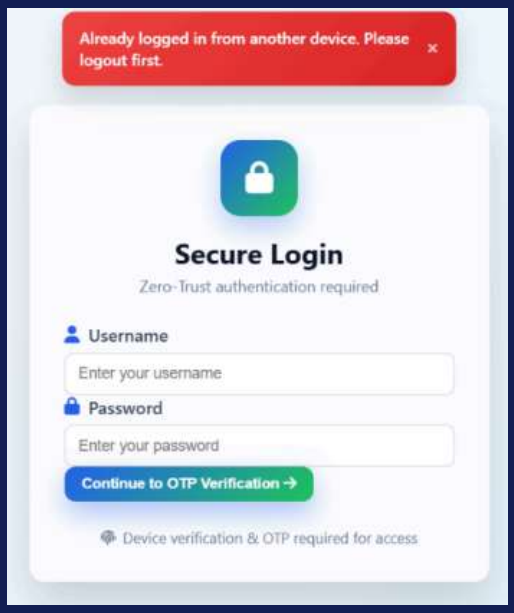

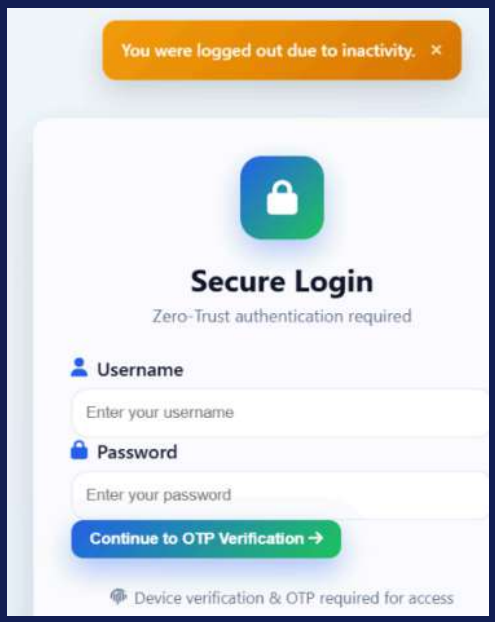

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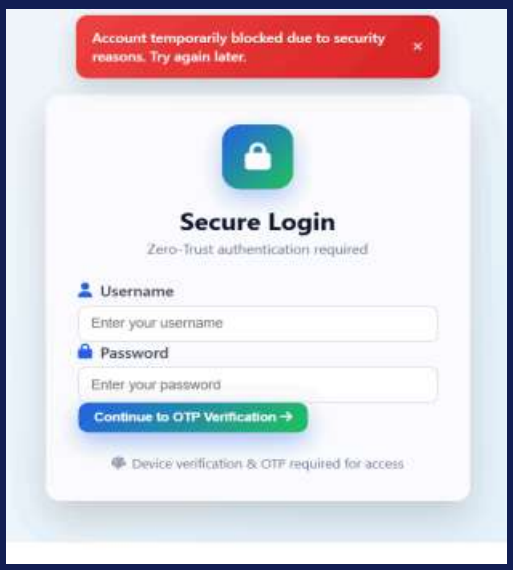

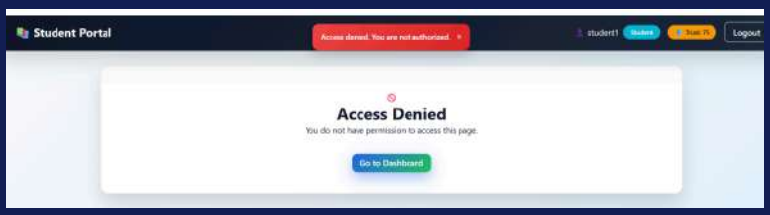

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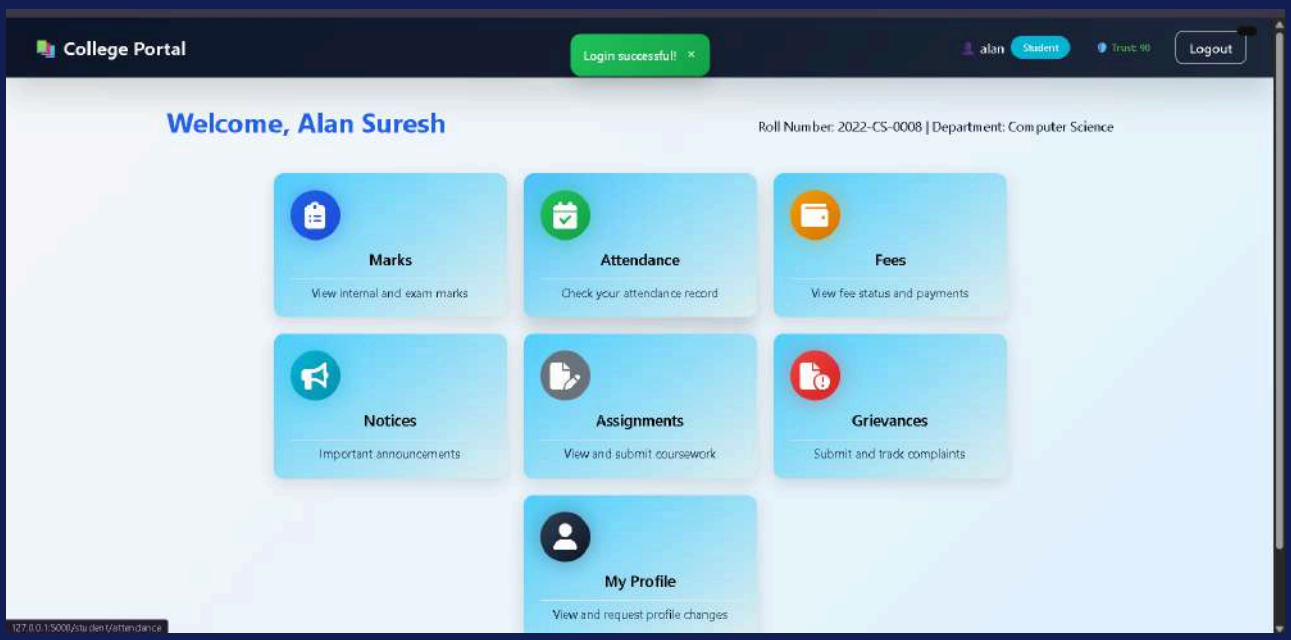

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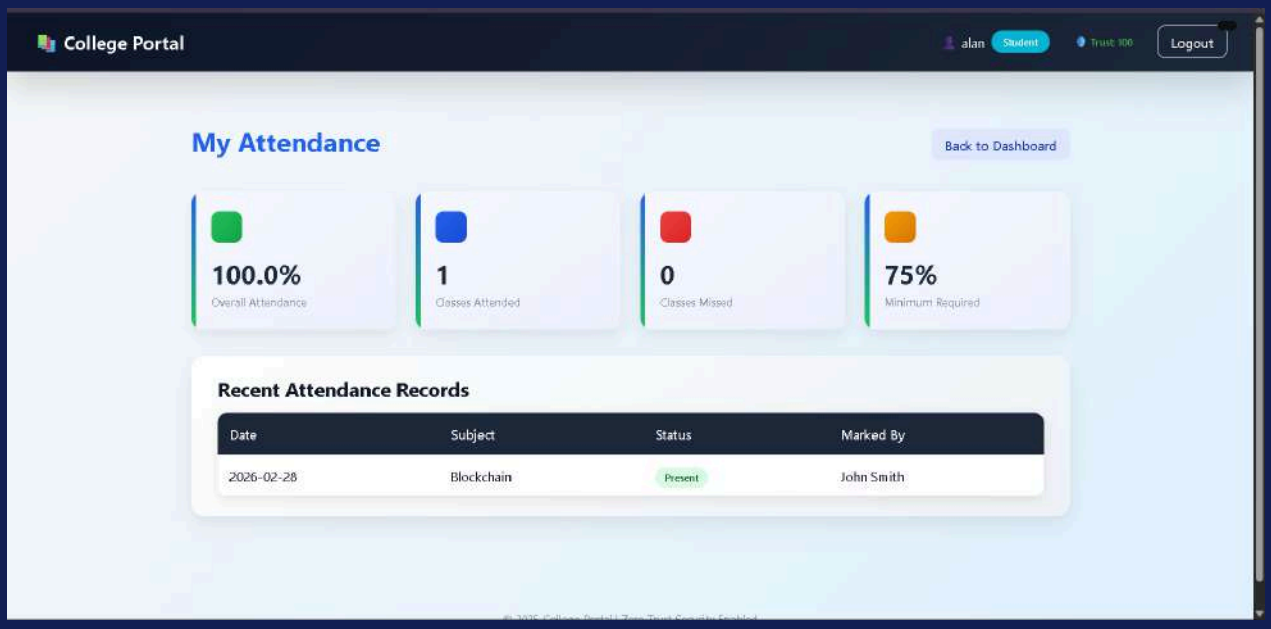
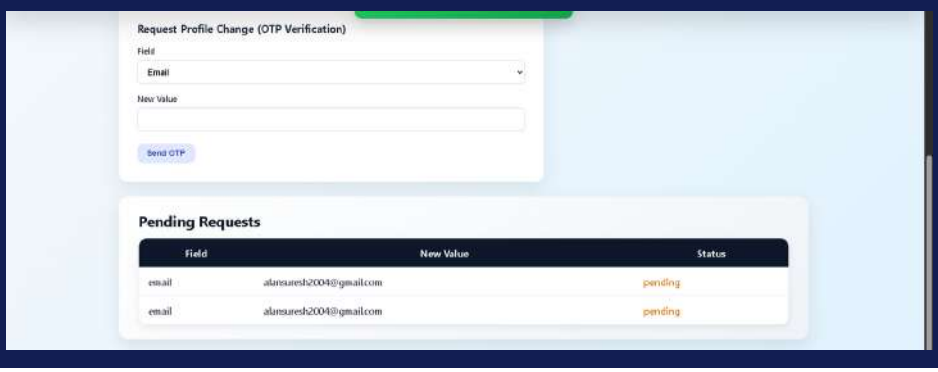


Results



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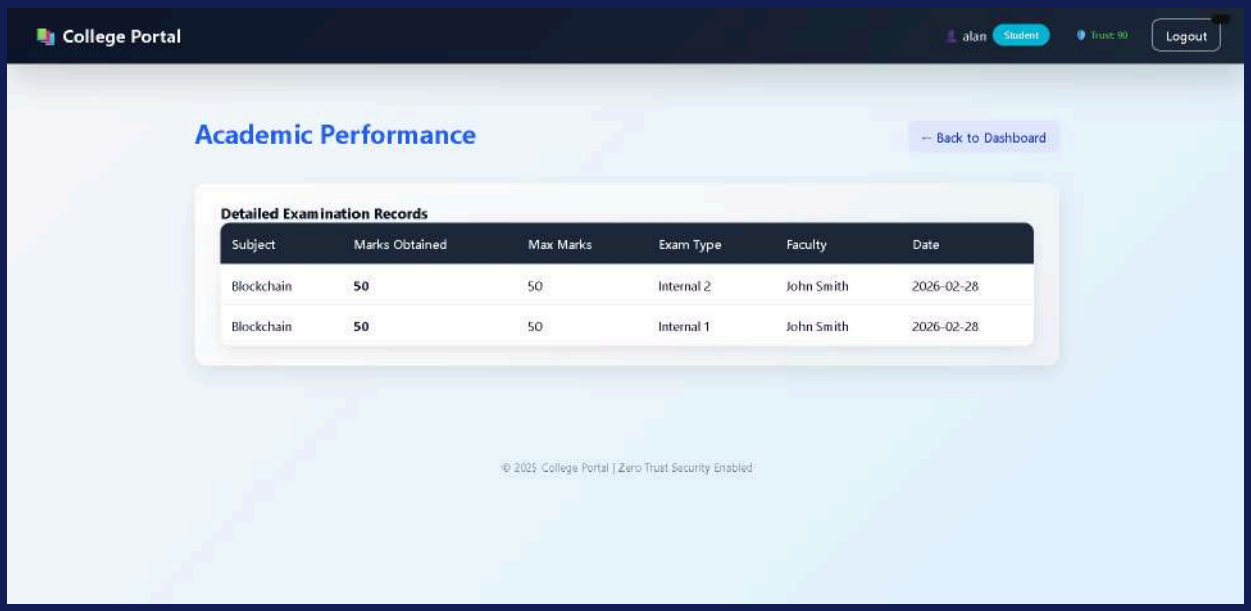

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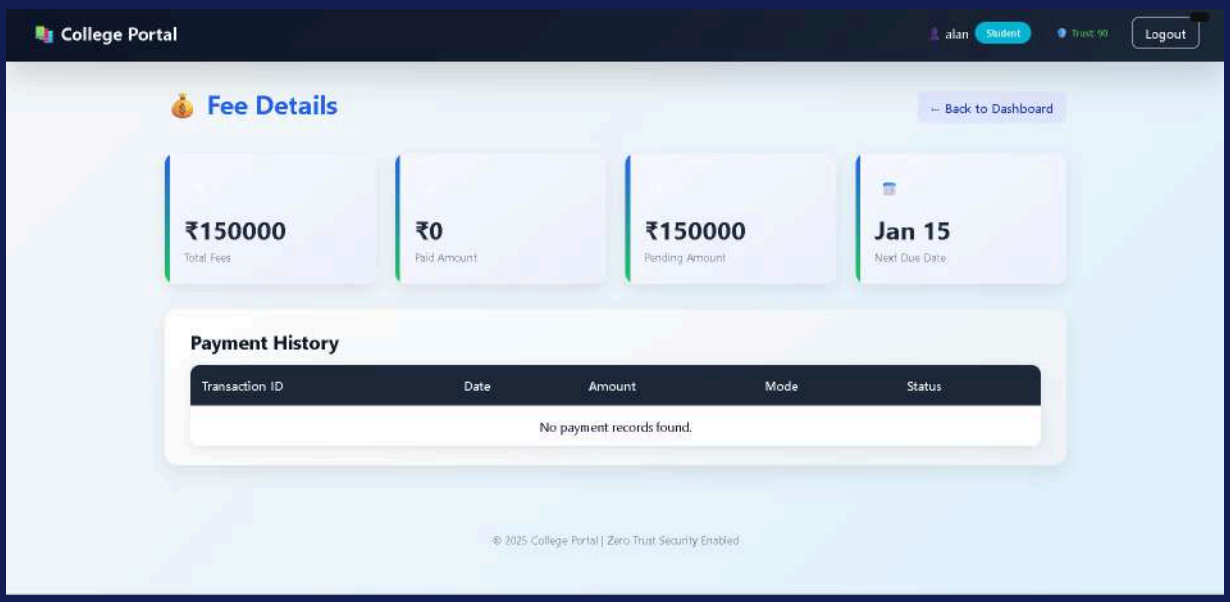

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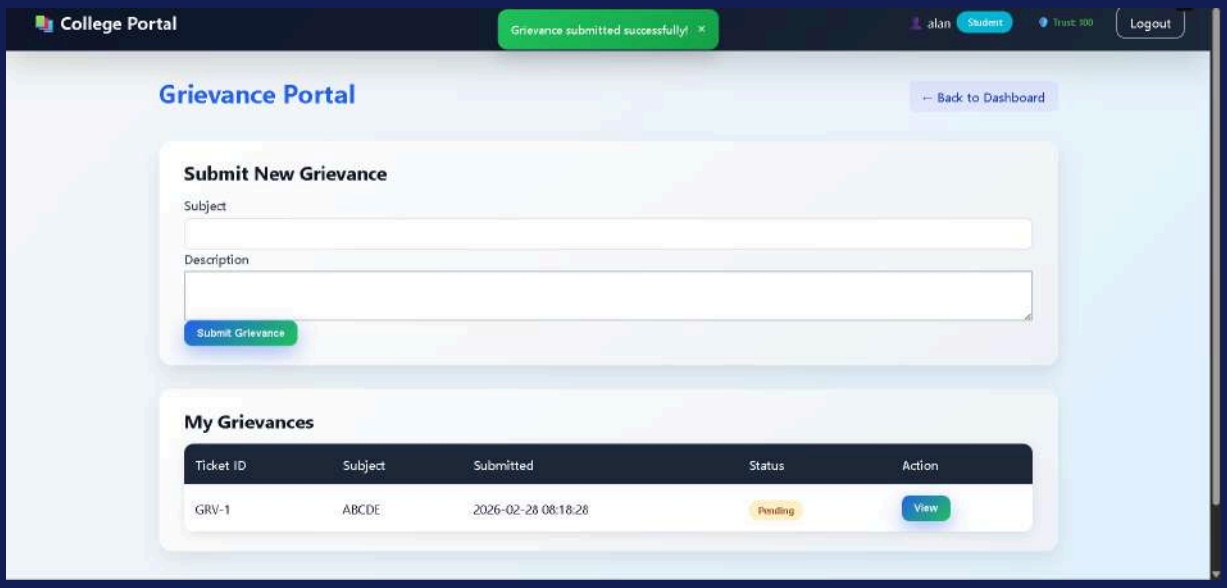

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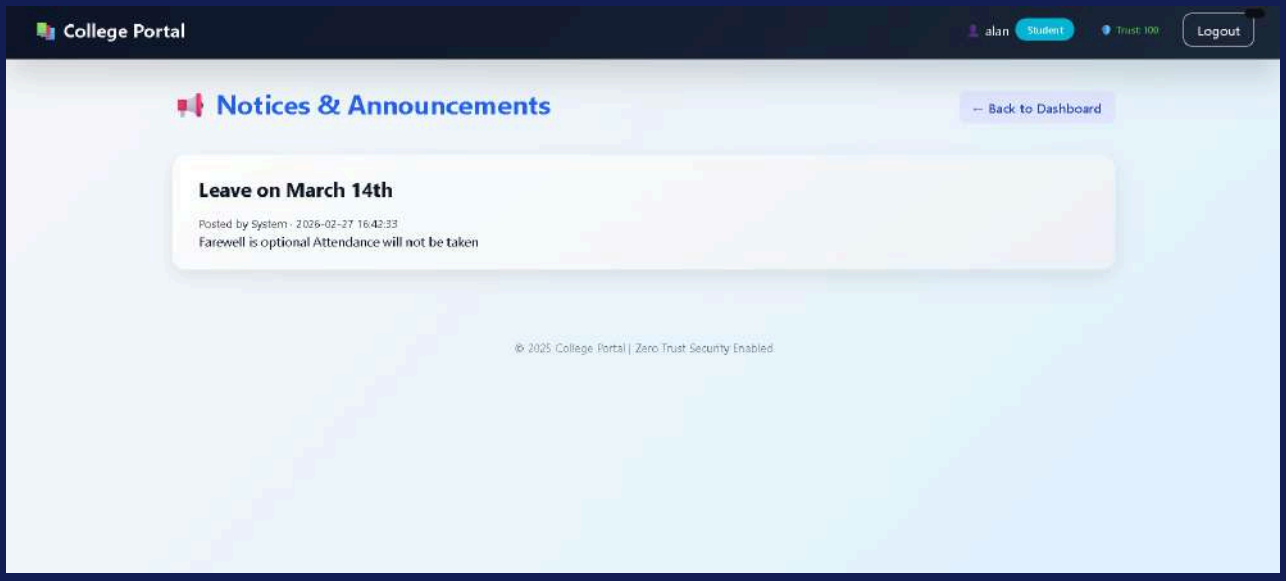

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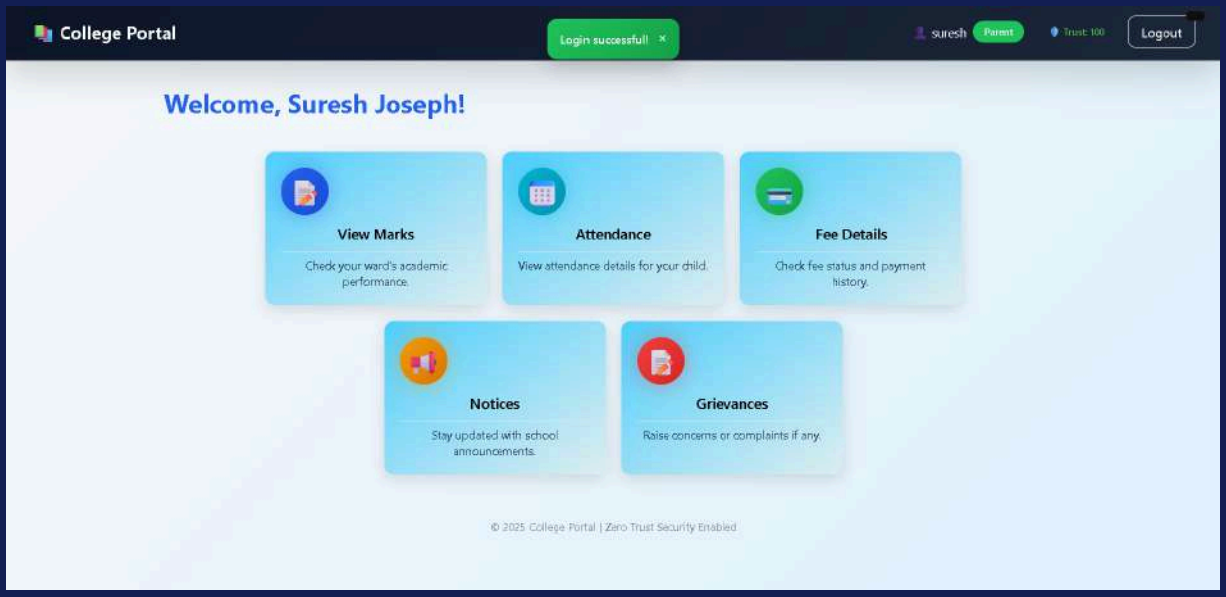

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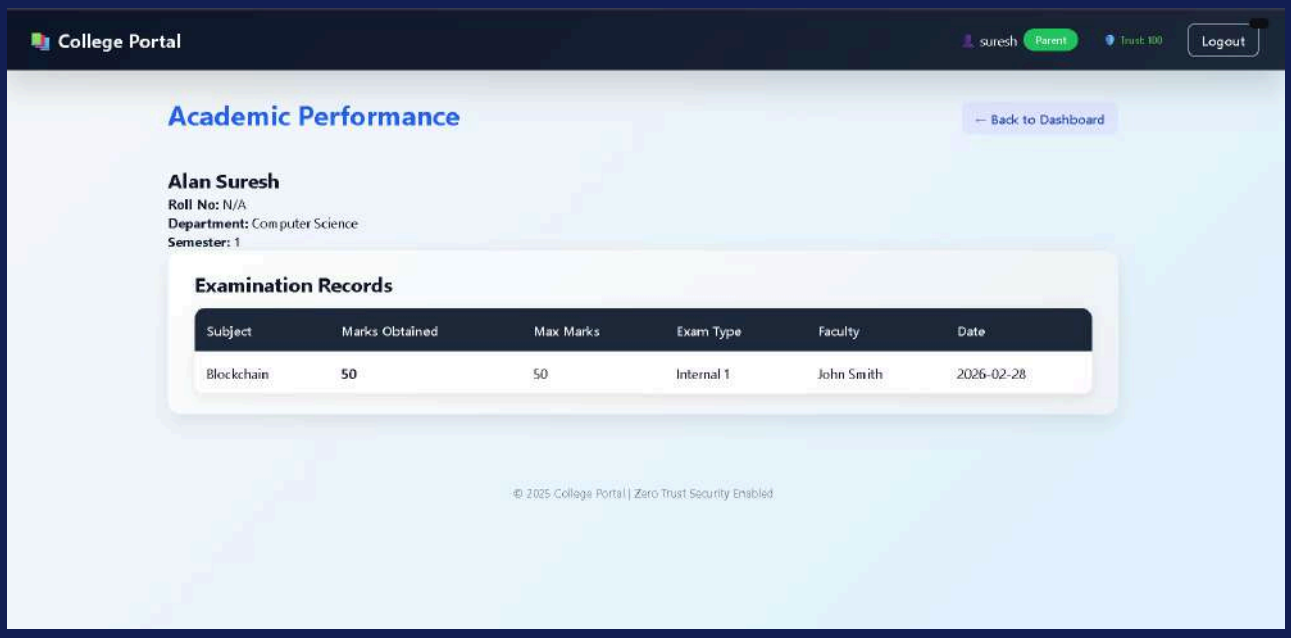

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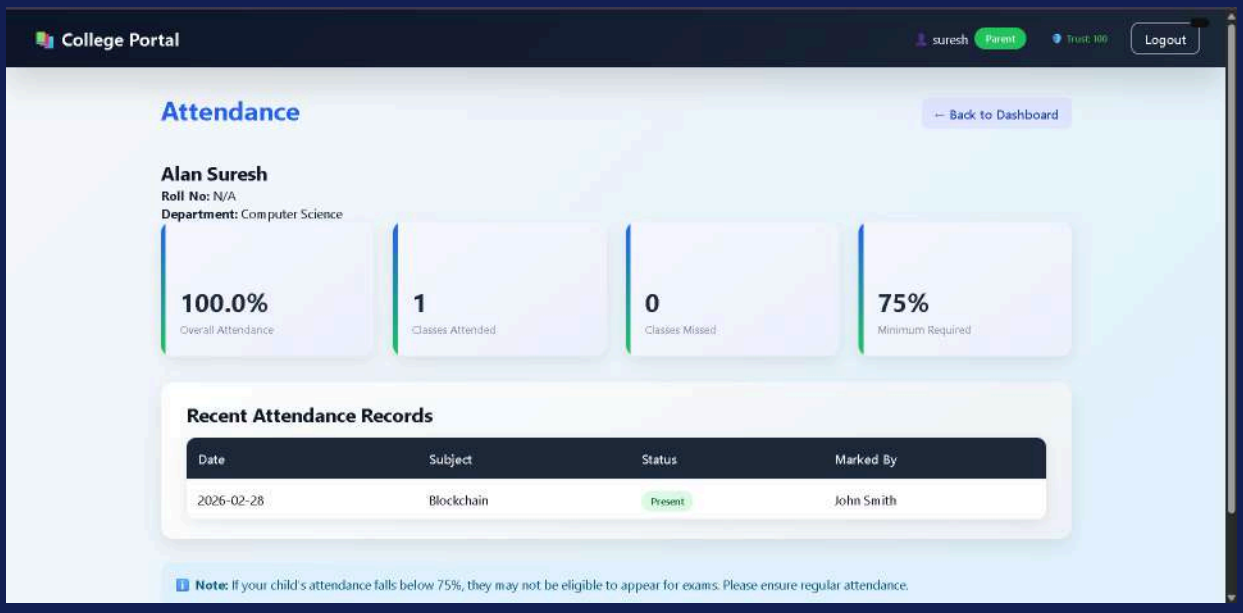

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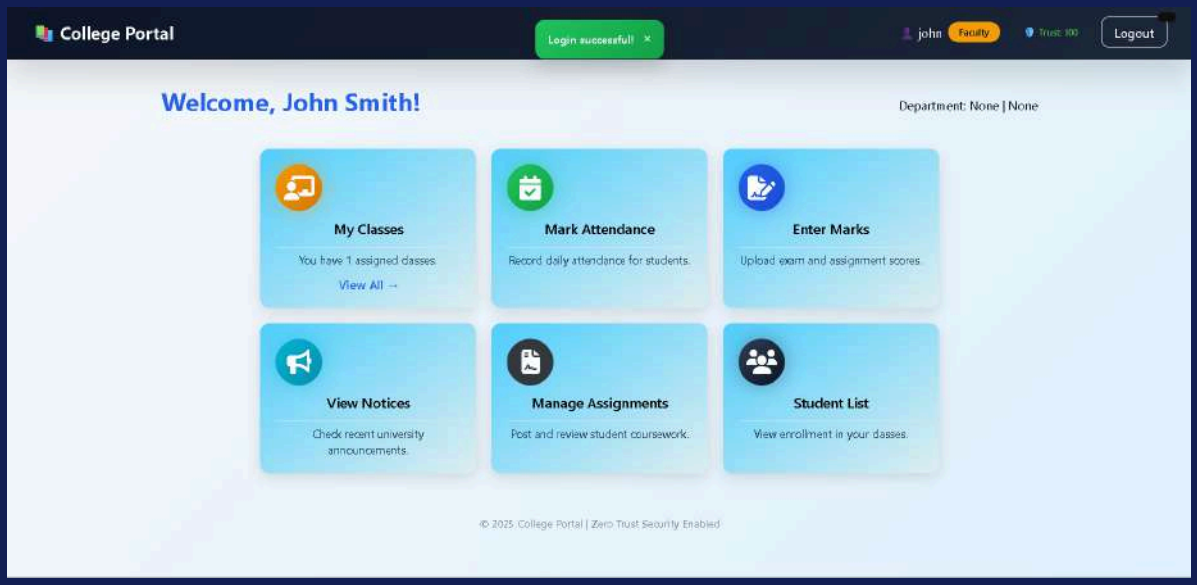

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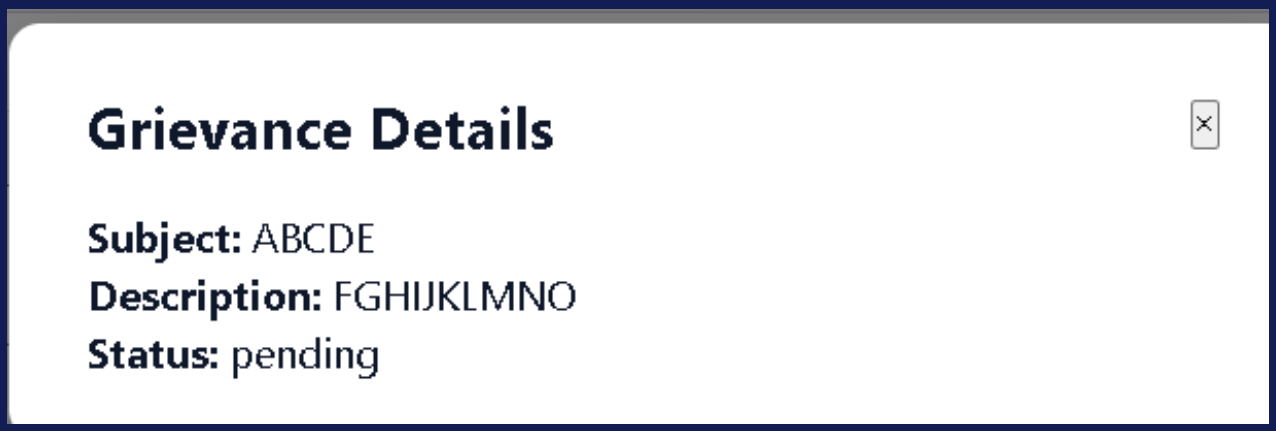

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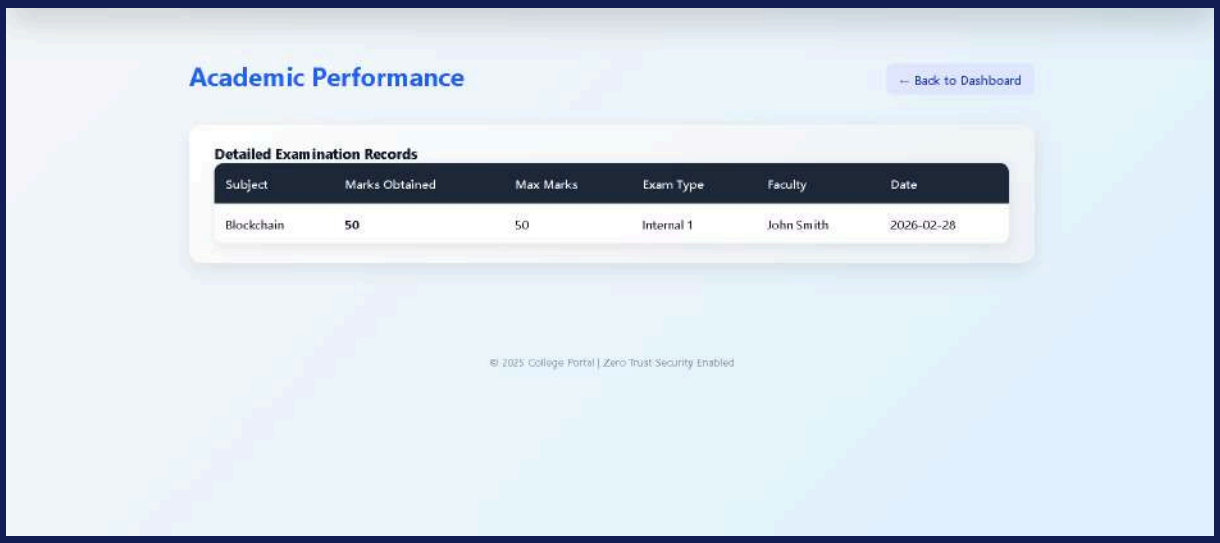

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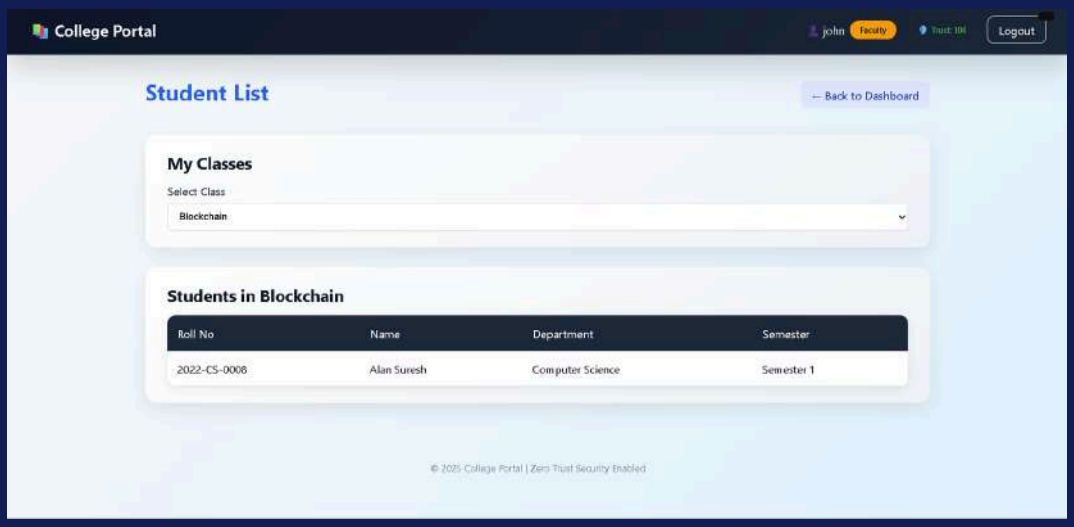

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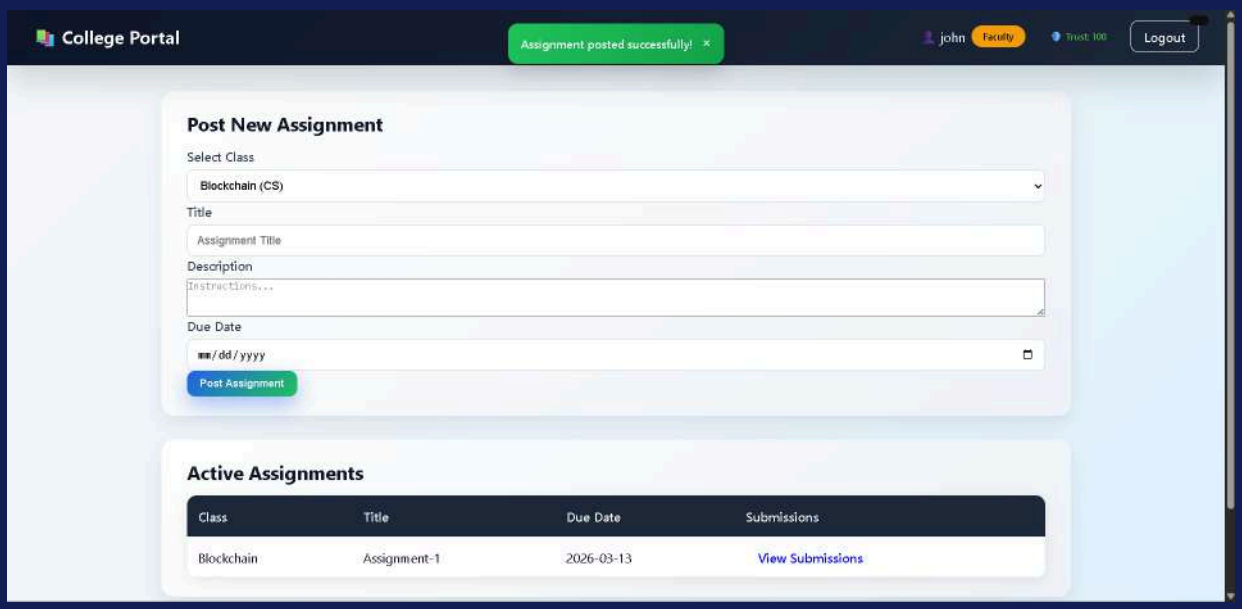

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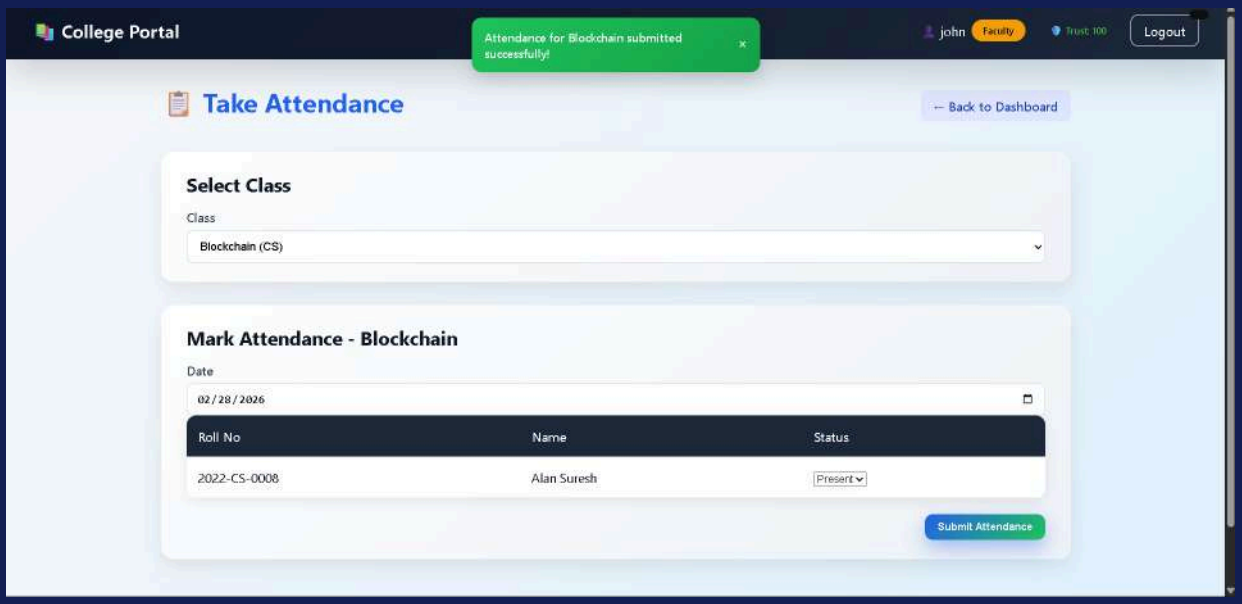

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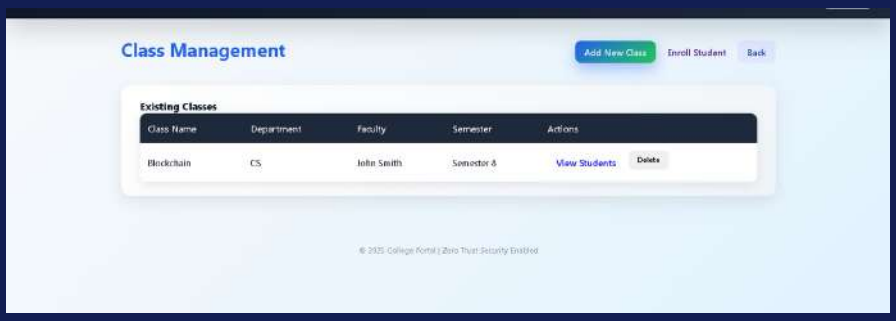

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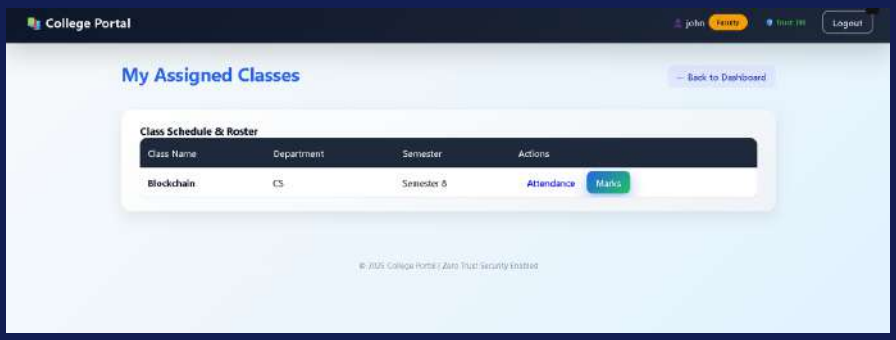

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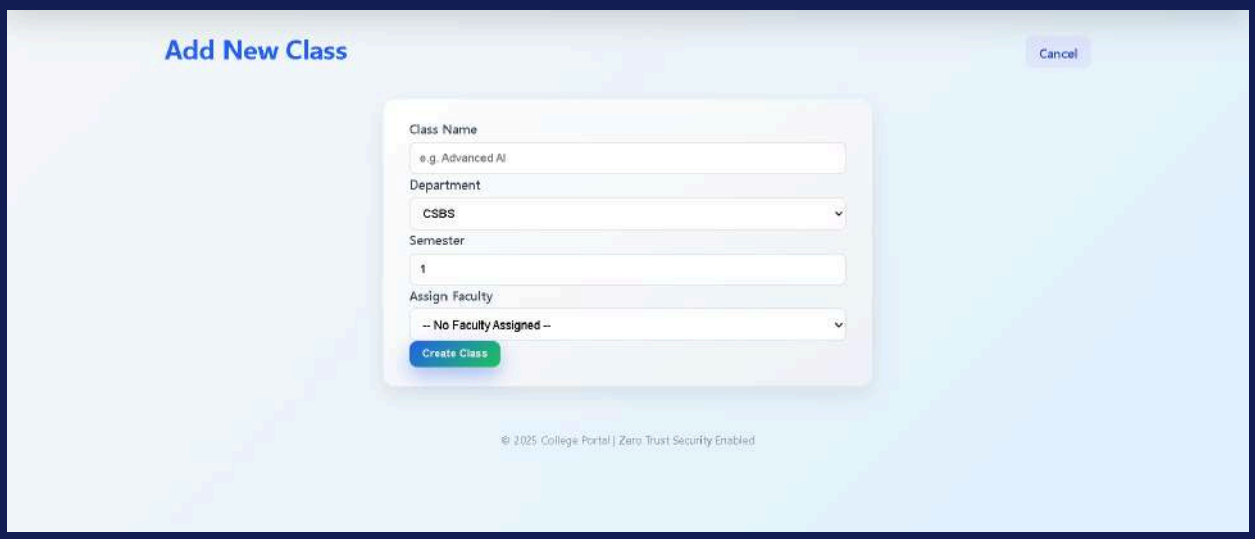

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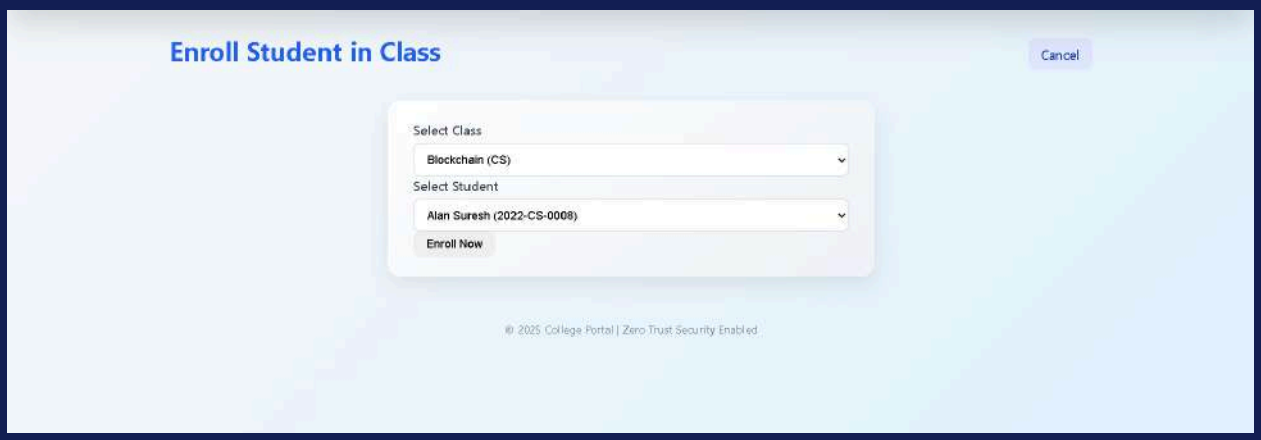

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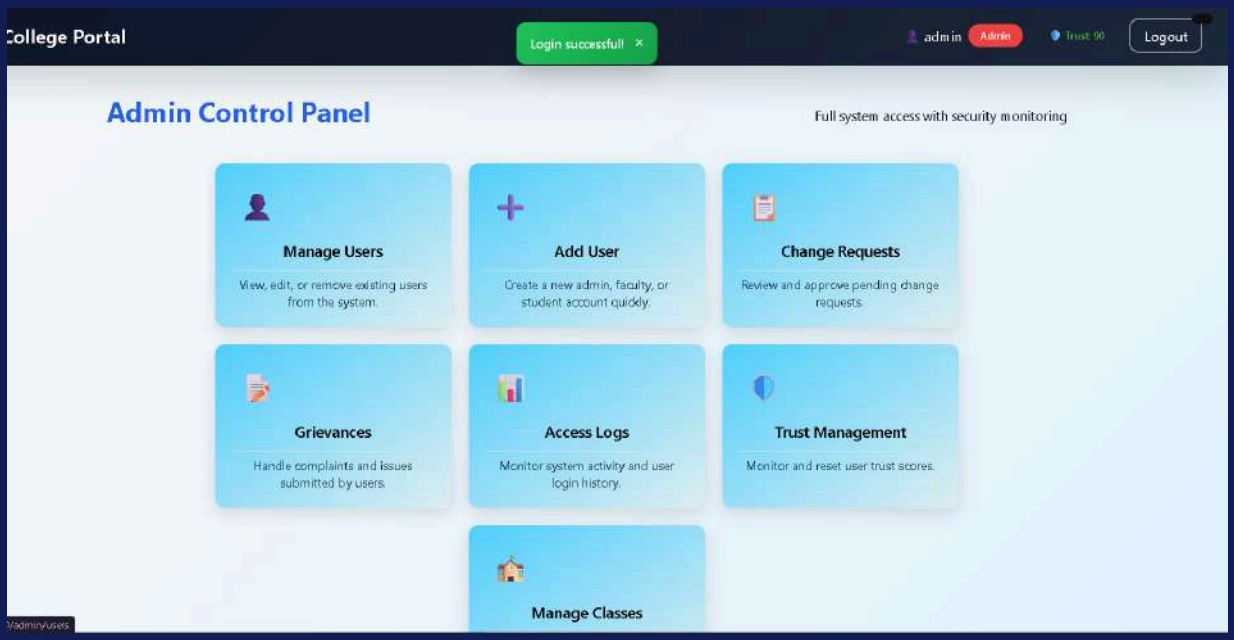

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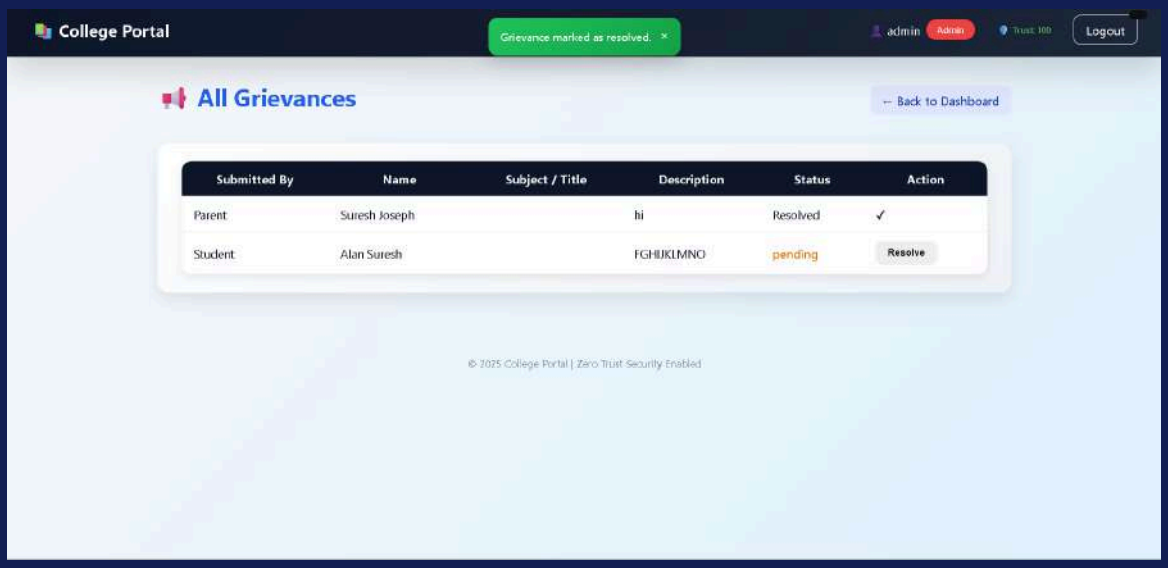
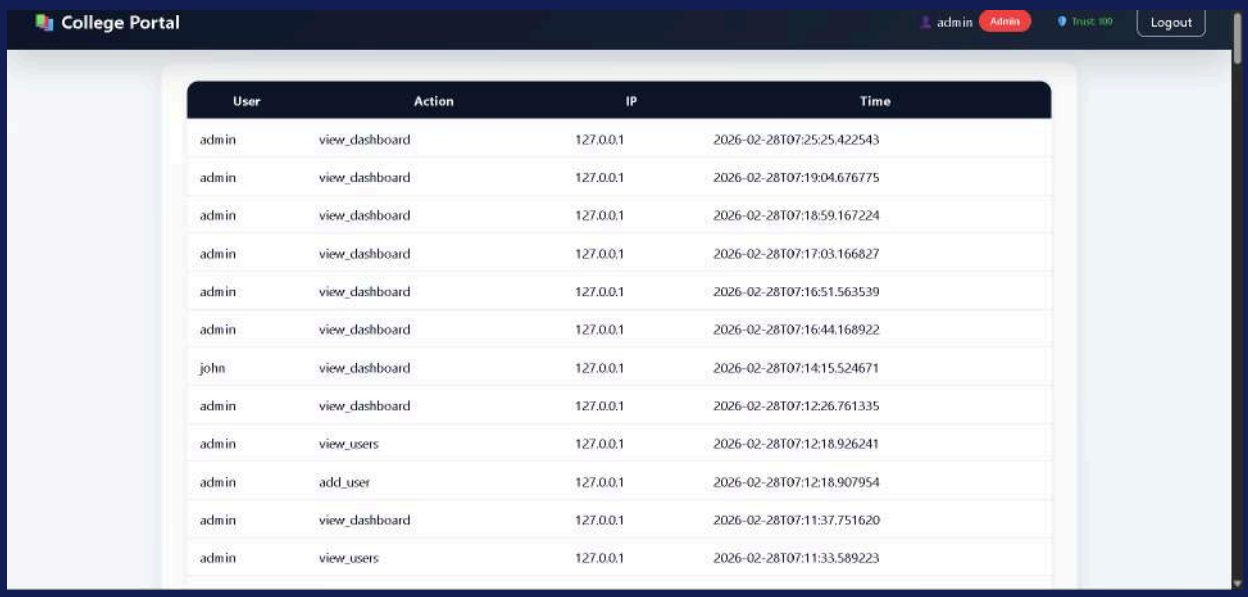


Results

Submitted By	Name	Subject / Title	Description	Status	Action
Parent	Suresh Joseph		hi	Pending	Resolve
Student	Alan Suresh		FGHIJKLMNO	pending	Resolve

Results



User	Action	IP	Time
admin	view_dashboard	127.0.0.1	2026-02-28T07:25:25.422543
admin	view_dashboard	127.0.0.1	2026-02-28T07:19:04.676775
admin	view_dashboard	127.0.0.1	2026-02-28T07:18:59.167224
admin	view_dashboard	127.0.0.1	2026-02-28T07:17:03.166827
admin	view_dashboard	127.0.0.1	2026-02-28T07:16:51.563539
admin	view_dashboard	127.0.0.1	2026-02-28T07:16:44.168922
john	view_dashboard	127.0.0.1	2026-02-28T07:14:15.524671
admin	view_dashboard	127.0.0.1	2026-02-28T07:12:26.761335
admin	view_users	127.0.0.1	2026-02-28T07:12:18.926241
admin	add_user	127.0.0.1	2026-02-28T07:12:18.907954
admin	view_dashboard	127.0.0.1	2026-02-28T07:11:37.751620
admin	view_users	127.0.0.1	2026-02-28T07:11:33.589223

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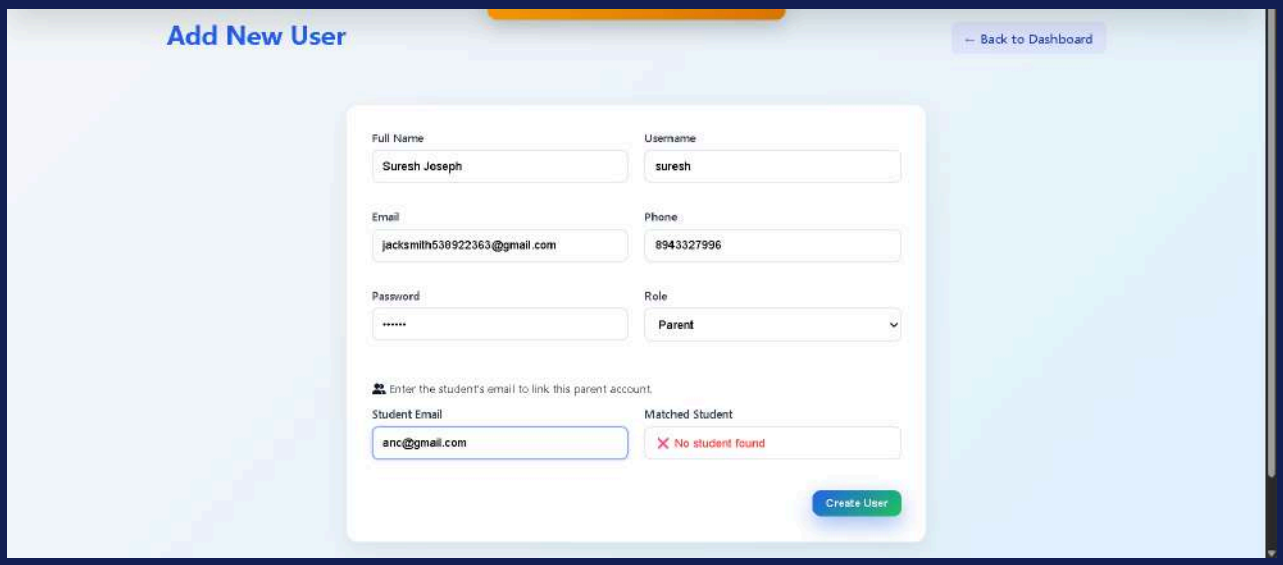

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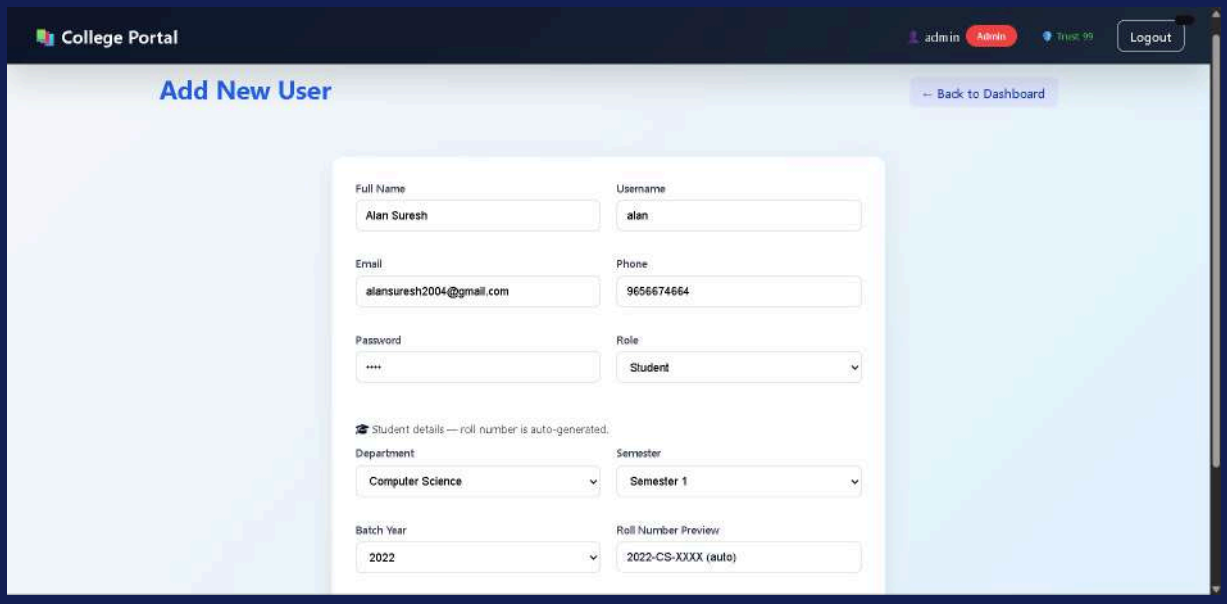

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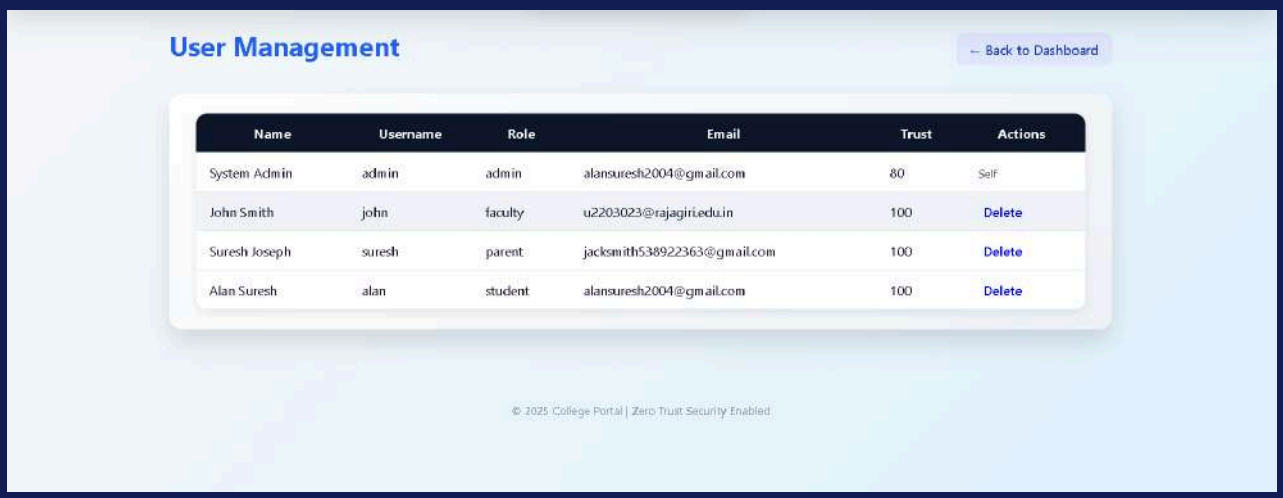

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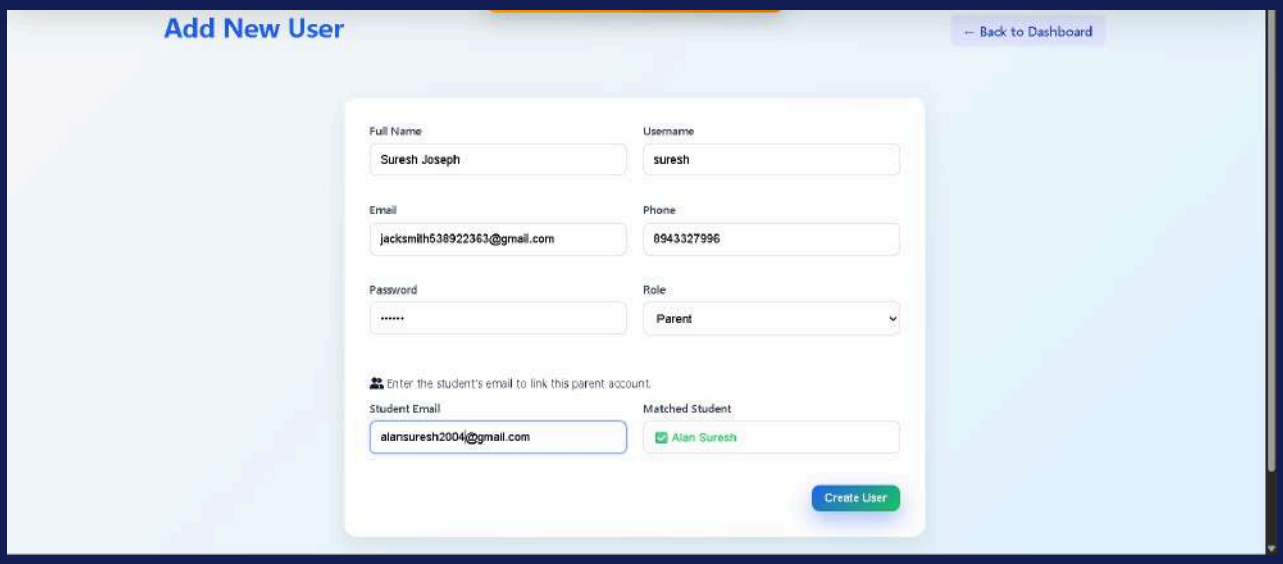


Assumptions

- Users possess valid credentials before accessing the system
- All access occurs through the secure VPN tunnel
- Internet connectivity is stable for OTP and authentication flows
- Database remains available and securely configured
- Institutional environment follows defined academic role structures (Student, Parent, Faculty, Admin)

Work Breakdown and Responsibilities

- Alan Suresh: Designed and implemented the Secure VPN Tunnel including tunnel establishment and teardown mechanisms, secure connection handling, authentication integration, encrypted communication flow management, and VPN activity logging.

- Aparna V Sunil: Developed the Secure VPN cryptographic layer including RSA-based key exchange, AES encryption and decryption mechanisms, secure session key management, data integrity validation, and cryptographic security logging.

- Abhirami C U: Designed and implemented the Zero Trust Policy Engine including trust score evaluation, device fingerprinting, concurrent session restriction, dynamic access control, session monitoring, OTP/MFA authentication, centralized audit logging, and integration of all four portals into a unified architecture.

- Aryasree Nambiar: Designed and developed the Zero Trust–enabled Student, Parent, Faculty, and Admin portals including secure login mechanisms, role-based access control (RBAC), academic management features, trust score integration, session timeout handling, and comprehensive audit logging.

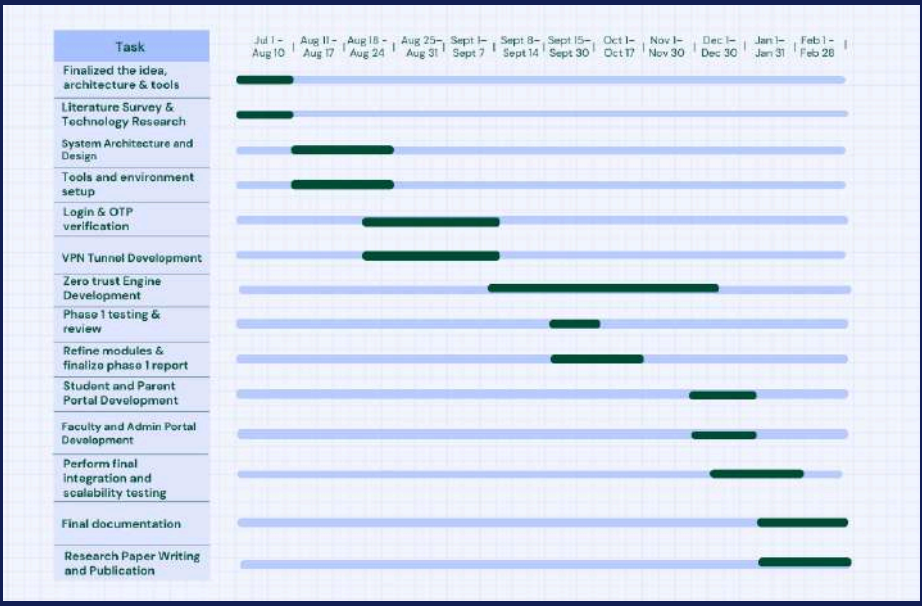
Risks and Challenges

- Potential vulnerabilities in custom VPN or cryptographic implementation
- Misconfiguration of RBAC policies leading to unauthorized access
- JWT token misuse or session hijacking if improperly secured
- OTP delivery delays affecting login experience
- Database breaches or improper data protection
- Scalability challenges with increasing concurrent users
- Integration complexity between VPN, authentication, and policy engine
- Performance overhead due to continuous trust evaluation and encryption

Hardware and Software Requirements

- Hardware Requirements
  - Laptop/Desktop (8 GB RAM or higher recommended)
  - Stable internet connection
  - Server or cloud instance for deployment
  - Secure campus network or cloud environment
- Software Requirements
  - Python (Flask Framework)
  - SQLite Database
  - HTML, CSS, Bootstrap (Frontend)
  - JWT & Cryptography Libraries (AES + RSA)
  - VS Code / IDE for development
  - GitHub for version control

Gantt Chart



Conclusion

- Successfully implemented a Zero Trust–based Academic Access Management System ensuring secure and role-driven access.
- Integrated strong authentication, encryption, and VPN security to protect academic data.
- Enforced RBAC, trust score evaluation, and continuous verification for controlled access.
- Enabled secure microsegmented portals with centralized management and audit logging.
- Demonstrated how Zero Trust principles enhance cybersecurity while maintaining usability and efficient administration.

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THANK YOU

Appendix B:
Vision, Mission, Programme Outcomes and Course Outcomes

Vision, Mission, Programme Outcomes and Course Outcomes

Institute Vision

To evolve into a premier technological institution, moulding eminent professionals with creative minds, innovative ideas and sound practical skill, and to shape a future where technology works for the enrichment of mankind.

Institute Mission

To impart state-of-the-art knowledge to individuals in various technological disciplines and to inculcate in them a high degree of social consciousness and human values, thereby enabling them to face the challenges of life with courage and conviction.

Department Vision

To become a centre of excellence in Computer Science and Engineering, moulding professionals catering to the research and professional needs of national and international organizations.

Department Mission

To inspire and nurture students, with up-to-date knowledge in Computer Science and Engineering, ethics, team spirit, leadership abilities, innovation and creativity to come out with solutions meeting societal needs.

Programme Outcomes (PO)

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis amp; interpretation of data to provide valid conclusions.

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering amp; IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national amp; international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to

one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

Programme Specific Outcomes (PSO)

A graduate of the Computer Science and Engineering Program will demonstrate:

PSO1: Computer Science Specific Skills: The ability to identify, analyze and design solutions for complex engineering problems in multidisciplinary areas by understanding the core principles and concepts of computer science and thereby engage in national grand challenges.

PSO2: Programming and Software Development Skills: The ability to acquire programming efficiency by designing algorithms and applying standard practices in software project development to deliver quality software products meeting the demands of the industry.

PSO3: Professional Skills: The ability to apply the fundamentals of computer science in competitive research and to develop innovative products to meet the societal needs thereby evolving as an eminent researcher and entrepreneur.

Course Outcomes (CO)

After the completion of the course the student will be able to:

CO1: Model and solve real world problems by applying knowledge across domains (Cognitive knowledge level: Apply).

C02: Develop products, processes or technologies for sustainable and socially relevant applications (Cognitive knowledge level: Apply).

C03: Function effectively as an individual and as a leader in diverse teams and to comprehend and execute designated tasks (Cognitive knowledge level: Apply).

CO4: Plan and execute tasks utilizing available resources within timelines, following ethical and professional norms (Cognitive knowledge level: Apply).

CO5: Identify technology/research gaps and propose innovative/creative solutions (Cognitive knowledge level: Analyze).

CO6: Organize and communicate technical and scientific findings effectively in written and oral forms (Cognitive knowledge level: Apply).

Appendix C: CO-PO-PSO Mapping

CO-PO AND CO-PSO MAPPING

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3
CO1	2	2	2	1		2	1	1	1	1	1	2	2	2
CO2	2	2	2		2	3	2	1	1	1	1	2	2	
CO3								3	2	2	1			3
CO4		1	2		3	2	3	2	2	3	1		2	2
CO5	2	3	3	1	2					1	3	3		
CO6			2			2	2	2	3	1	1			3

Table 5.1: CO-PO-PSO Mapping

JUSTIFICATIONS FOR CO-PO MAPPING

Mapping	**Level**	**Justification**
101003/ CS822U.CO1-PO1	Medium	Applies engineering knowledge, mathematics, and computing fundamentals to model and solve real-world problems.
101003/ CS822U.CO1-PO2	Medium	Requires identification, formulation and analysis of engineering problems before developing solutions.
101003/ CS822U.CO1-PO3	Medium	Involves designing systems or solutions to address identified engineering needs.
101003/ CS822U.CO1-PO4	Low	Limited investigation or experimentation may be required to validate the developed models or solutions.
101003/ CS822U.CO1-PO6	Medium	Solutions developed may influence societal, environmental, or sustainability aspects of engineering applications.
101003/ CS822U.CO1-PO7	Medium	Encourages consideration of ethical and professional responsibilities while developing solutions.
101003/ CS822U.CO1-PO8	Low	Some level of teamwork may be required while working on modeling or project tasks.

Mapping	Level	Justification
101003/ CS822U.CO1-PO9	Low	Requires communication of results and findings through reports or presentations.
101003/ CS822U.CO1-PO10	Low	Involves basic planning and execution of tasks while implementing solutions.
101003/ CS822U.CO1-PO11	Low	Encourages independent learning and adapting new knowledge to solve real-world problems.
101003/ CS822U.CO1-PSO1	Medium	Uses core computer science principles to analyze problems and design appropriate technical solutions.
101003/ CS822U.CO1-PSO2	Medium	Involves algorithm design and application of software development practices to implement solutions.
101003/ CS822U.CO1-PSO3	Medium	Applies computer science fundamentals in research or innovative problem solving for societal needs.
101003/ CS822U.CO2-PO1	Medium	Utilizes engineering knowledge to design and develop products or systems.
101003/ CS822U.CO2-PO2	Medium	Requires analysis of user needs and technological feasibility before development.
101003/ CS822U.CO2-PO3	Medium	Focuses on design and development of solutions addressing real-world requirements.
101003/ CS822U.CO2-PO5	Medium	Requires use of modern engineering and IT tools for system or product development.
101003/ CS822U.CO2-PO6	High	Emphasizes sustainability and societal impact during solution development.

Mapping	Level	Justification
101003/ CS822U.CO2-PO7	Medium	Ethical and professional responsibilities are considered when developing socially relevant technologies.
101003/ CS822U.CO2-PO8	Low	Development activities may require collaboration among team members.
101003/ CS822U.CO2-PO9	Low	Involves communicating design ideas and development outcomes.
101003/ CS822U.CO2-PO10	Low	Requires basic planning and coordination during development tasks.
101003/ CS822U.CO2-PO11	Low	Encourages continuous learning of emerging technologies while building solutions.
101003/ CS822U.CO2-PSO1	Medium	Applies computer science fundamentals to design technically feasible solutions.
101003/ CS822U.CO2-PSO2	Medium	Uses programming and software development practices to implement solutions.
101003/ CS822U.CO3-PO8	High	Strongly relates to functioning effectively as a member or leader in multidisciplinary teams.
101003/ CS822U.CO3-PO9	Medium	Requires effective communication among team members during collaboration.
101003/ CS822U.CO3-PO10	Medium	Team leadership involves project coordination and task management.
101003/ CS822U.CO3-PO11	Low	Team interaction supports learning from peers and adapting new approaches.

Mapping	Level	Justification
101003/ CS822U.CO3-PSO3	High	Professional collaboration and teamwork support innovative computer science solutions.
101003/ CS822U.CO4-PO2	Low	Requires analysis and planning before executing project tasks.
101003/ CS822U.CO4-PO3	Medium	Involves designing and implementing project solutions based on planned activities.
101003/ CS822U.CO4-PO5	High	Requires effective use of modern tools and resources during project execution.
101003/ CS822U.CO4-PO6	Medium	Considers societal and environmental impact while implementing solutions.
101003/ CS822U.CO4-PO7	High	Emphasizes adherence to ethical practices and professional responsibilities.
101003/ CS822U.CO4-PO8	Medium	Task execution may involve teamwork and coordination among team members.
101003/ CS822U.CO4-PO9	Medium	Requires communication of progress, outcomes and documentation.
101003/ CS822U.CO4-PO10	High	Focuses on project management including scheduling, resource allocation and task execution.
101003/ CS822U.CO4-PO11	Low	Encourages adaptability and continuous learning during project implementation.
101003/ CS822U.CO4-PSO2	Medium	Uses software development practices to execute and manage project tasks.

Mapping	Level	Justification
101003/ CS822U.CO4-PSO3	Medium	Applies computer science knowledge in practical project development environments.
101003/ CS822U.CO5-PO1	Medium	Uses engineering knowledge to understand technological limitations and opportunities.
101003/ CS822U.CO5-PO2	High	Requires deep problem analysis and identification of research gaps.
101003/ CS822U.CO5-PO3	High	Proposes innovative design solutions to address identified gaps.
101003/ CS822U.CO5-PO4	Low	May involve limited investigation or validation of proposed ideas.
101003/ CS822U.CO5-PO5	Medium	Uses appropriate tools and techniques to analyze and develop innovative solutions.
101003/ CS822U.CO5-PO10	Low	Requires minimal planning while proposing new solution approaches.
101003/ CS822U.CO5-PO11	High	Strongly promotes independent learning and exploration of emerging technologies.
101003/ CS822U.CO5-PSO1	High	Applies computer science fundamentals to identify and solve complex technical problems.
101003/ CS822U.CO6-PO3	Medium	Requires presentation of design concepts and developed solutions.
101003/ CS822U.CO6-PO6	Medium	Communication may include discussion of societal or environmental implications.

Mapping	Level	Justification
101003/ CS822U.CO6-PO7	Medium	Requires responsible and ethical communication of technical results.
101003/ CS822U.CO6-PO8	Medium	Communication supports collaboration within team environments.
101003/ CS822U.CO6-PO9	High	Strongly relates to effective written and oral communication of technical findings.
101003/ CS822U.CO6-PO10	Low	Requires structured reporting and documentation in project management contexts.
101003/ CS822U.CO6-PO11	Low	Encourages continuous improvement in communication and presentation skills.
101003/ CS822U.CO6-PSO3	High	Communication of innovative computer science solutions and research outcomes.